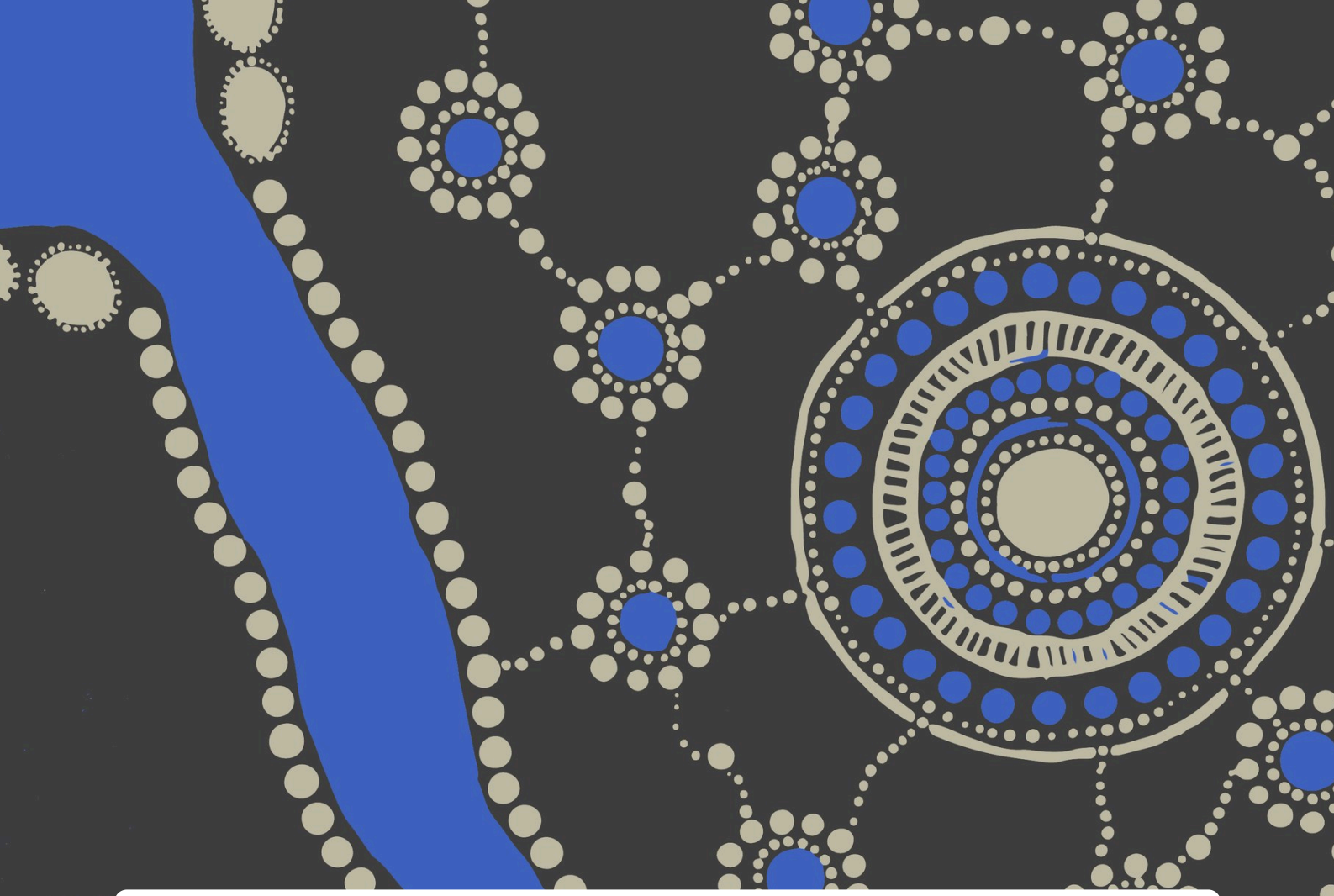


KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics Standard 2
Year 12

CHAPTER 1 · KEY IDEAS AND REVIEW

Ratios and rates

Year 12 Standard 2 • Simplified review workbook

NSW Mathematics Standard 11–12 Syllabus (2024) • Outcomes MST-12-S2-05, MAO-WM-01

Name _____ Class _____ Teacher _____

Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

About this workbook

This workbook is a simplified version of the **Chapter 1 key ideas and review** from your textbook, *CambridgeMATHS Mathematics Standard 2 Year 12*. There is one lesson for each key idea, and each lesson has three parts:

Key idea: the idea in plain words, with the *I can* statements from the textbook checklist.

Worked examples: every example is fully worked, with the reason for each line next to it. The answer is in a blue box.

Your turn: **multiple choice**, **short answer** and **extended response** questions, like the textbook review. Each question tells you where to look.

The workbook ends with a **chapter review** set out the same way, and answers to every question.

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WORKING ON YOUR OWN?

1. Read the key idea and look at the picture.
2. Read a worked example one line at a time. Cover the next line: can you say what comes next?
3. Try the questions that point to that example. Write every step in the box.
4. Check the answers at the back, fix any mistakes, then rate yourself on the checklist.

Syllabus outcomes

MST-12-S2-05	solves practical problems involving ratios and rates
MAO-WM-01	develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly

Formulas you'll use (some are on the HSC reference sheet)

Trapezoidal rule: $A \approx \frac{h}{2}(d_f + d_l)$

Speed: $S = \frac{D}{T}$

Volume: $V = Ah$, $1\ \text{m}^3 = 1000\ \text{L}$

Fuel used (L) $= \frac{\text{distance (km)}}{100} \times \text{rate (L/100 km)}$

Energy (kWh) = power (kW) × time (h)

Acknowledgement of Country

We acknowledge the Traditional Custodians of the land on which we learn, the Bundjalung people, and pay our respects to Elders past and present. The artwork on the cover and throughout this workbook is our school vision statement, created by local Indigenous artist Marni Tuala.

You will need

A calculator, a ruler and a pencil. Show your working in every box: in the HSC, working earns marks even when the final answer is wrong.

Key ideas Chapter 1 at a glance

A simplified version of the textbook's *Key ideas and chapter summary*. Each key idea has its own lesson.

			EXAMPLE
1	Ratio	Compares amounts of the **same kind**, in a set order. Multiply or divide every part by the same number to get an **equivalent ratio**.	5 : 2 is not the same as 2 : 5 5 : 2 = 10 : 4 = 15 : 6
2	Capture–recapture	Estimates the size of an animal population. Tag some animals, release them, then catch a second sample.	$\frac{\text{tagged in 2nd sample}}{\text{size of 2nd sample}} = \frac{\text{tagged at first}}{\text{population}}$
3	Dividing a quantity in a ratio	**1** Add the parts (2 + 3 = 5). **2** Write each part over the total. **3** Multiply the quantity by each fraction.	Share \$40 in the ratio 2 : 3. $\frac{2}{5} \times \$40 = \$16$, $\frac{3}{5} \times \$40 = \$24$
4	Scale drawings	Scale = drawing length : actual length. A scale can have units (1 cm to 1 m) or no units (1 : 100).	Scale 1 : 100. 3 cm on the plan → real length = 3 × 100 = 300 cm = 3 m
5	Plans and elevations	A **plan** is the view from the top. An **elevation** is the view from one side, such as the front or the side.	plan = bird's-eye view front elevation = view from the front
6	Perimeter and area of land	**Perimeter**: measure each side on the photo and multiply by the scale. **Area**: count grid squares to estimate, or use a formula. For one curved side, use the trapezoidal rule.	$A \approx \frac{h}{2}(d_f + d_l)$ h = distance between the two measurements, d_f = first, d_l = last
7	Volume of water	If the depth is the same everywhere, volume = area of the surface × depth. For rain, change millimetres to metres first.	$V = Ah$ (h = depth), 1 m^3 = 1000 L 18 mm = 0.018 m
8	Rates	Compares two amounts with **different units**.	km/h, \$/kg, L/100 km, \$/hour 25 mL/min = 25 × 60 mL/h = 1500 mL/h = 1.5 L/h
9	Unitary method	Divide to find the amount for **1**. Multiply to find the amount you need. For the best buy, compare prices for the same amount.	3 kg costs \$12 1 kg: \$12 ÷ 3 = \$4 → 5 kg: 5 × \$4 = \$20
10	Speed	Speed is a rate. On a distance–time graph, a steeper line is faster and a flat line means stopped.	$S = \frac{D}{T}$, $D = S \times T$
11	Energy	Energy (kWh) = power (kW) × time (hours). Running cost = energy × price per kWh.	1 kW = 1000 W, so 600 W = 0.6 kW 2 kW for 3 h = 6 kWh at \$0.30 per kWh: 6 × \$0.30 = \$1.80
12	Fuel consumption	A rate given in litres per 100 km (L/100 km) or km per litre.	Fuel used (L) = $\frac{\text{distance (km)}}{100} \times \text{rate}$ Rate = $\frac{\text{litres}}{\text{km}} \times 100$ 30 L for 400 km → 7.5 L/100 km

1 Ratios

Key idea: Ratio • Textbook 1A • MST-12-S2-05

Key idea Ratio

A **ratio** compares two or more amounts of the **same kind**, in a set order.



sand : cement = 5 : 2

- 5 buckets of sand to 2 buckets of cement is 5 : 2, read "5 to 2".
- **Order matters:** cement to sand is 2 : 5. A ratio has **no units**.
- **Equivalent ratios:** multiply or divide **every** part by the same number. 5 : 2 = 10 : 4 = 15 : 6

Simplest form

Divide every part by the **highest common factor** (HCF).

- **Finding the HCF:** try the factors of the smallest number, biggest first, until one divides into every part.
- **Decimals?** Multiply every part by 10 first.
- **Different units?** Make them the same: 40 cm : 2 m = 40 : 200 = 1 : 5.

Three ways to write a ratio

Form	How	8 : 10
Simplest	÷ by the HCF	4 : 5
1 : a	÷ by the **first** number	1 : 1.25
b : 1	÷ by the **second** number	0.8 : 1

Finding an unknown

Find two numbers in the same place. Divide to find what one was multiplied by. Do the **same** to the other part.

$3 : 4 = a : 20 \rightarrow 20 \div 4 = \mathbf{5}$, so $a = 3 \times \mathbf{5} = 15$

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* change a ratio to simplest form, to the form 1 : a and to the form b : 1. ○○○
- ▸ *I can* use a ratio to find an unknown amount. ○○○

WORKED EXAMPLE 1.1 Simplest form

Write each ratio in simplest form. (a) 12 : 18 (b) 10 : 15 : 30 (c) 4.8 : 1.6

	WORKING	WHY
(a)	12 : 18	The HCF of 12 and 18 is **6**.
	= 12 ÷ 6 : 18 ÷ 6	Divide **both** parts by 6.
	= 2 : 3	Only 1 divides into 2 and 3, so this is simplest form.
(b)	10 : 15 : 30	The HCF of 10, 15 and 30 is **5**.
	= 10 ÷ 5 : 15 ÷ 5 : 30 ÷ 5	Divide **every** part by 5.
	= 2 : 3 : 6	Only 1 divides into 2, 3 and 6.
(c)	4.8 : 1.6	There are decimals. Make whole numbers first.
	= 4.8 × 10 : 1.6 × 10	Multiply **both** parts by **10**.
	= 48 : 16	The HCF of 48 and 16 is **16**.
	= 48 ÷ 16 : 16 ÷ 16	Divide both parts by 16.
	= 3 : 1	Keep the 1. Write 3 : 1, not just 3.

WORKED EXAMPLE 1.2 — The forms $1 : a$ and $b : 1$

Write the ratio $15 : 12$ (a) in the form $1 : a$ (b) in the form $b : 1$.

	WORKING	WHY
(a)	$15 : 12$	To make the **first** part 1, divide both parts by the first number, 15.
	$= 15 \div 15 : 12 \div 15$	Use your calculator: $12 \div 15 = 0.8$
	$= \boxed{1 : 0.8}$	The first part is now 1.
(b)	$15 : 12$	To make the **second** part 1, divide both parts by the second number, 12.
	$= 15 \div 12 : 12 \div 12$	Use your calculator: $15 \div 12 = 1.25$
	$= \boxed{1.25 : 1}$	The second part is now 1.

WORKED EXAMPLE 1.3 — Finding an unknown

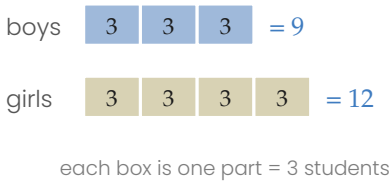
Find the value of the pronumeral. (a) $2 : 5 = a : 20$ (b) $3 : 4 = 15 : b$

	WORKING	WHY
(a)	$2 : 5 = a : 20$	Find two numbers in the same place: 5 and 20 are both **second** parts.
	$20 \div 5 = \mathbf{4}$	Divide to find what 5 was multiplied by ($5 \times 4 = 20$).
	$a = 2 \times \mathbf{4}$	Do the same to the first part.
	$\boxed{a = 8}$	Check: 8:20 simplifies to 2:5.
(b)	$3 : 4 = 15 : b$	3 and 15 are both **first** parts.
	$15 \div 3 = \mathbf{5}$	Divide to find what 3 was multiplied by ($3 \times 5 = 15$).
	$b = 4 \times \mathbf{5}$	Do the same to the second part.
	$\boxed{b = 20}$	Check: 15:20 simplifies to 3:4.

WORKED EXAMPLE 1.4 — Using a ratio to find an amount

The ratio of boys to girls in a class is 3 : 4. There are 12 girls. How many boys are in the class?

WORKING	WHY
boys : girls = 3 : 4	Write the ratio with labels so you know which part is which.
4 parts = 12 girls	Girls are the **second** part, so 4 parts = 12.
1 part = 12 ÷ 4 = 3	Find **one part** first.
boys = 3 parts = 3 × 3	Boys are the **first** part, 3 parts.
= 9 boys	Check: 9:12 simplifies to 3:4.



each box is one part = 3 students

YOUR TURN 1

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Which is equivalent to 2 : 5?
→ Key idea

A. 4 : 10 **B.** 5 : 2
C. 4 : 7 **D.** 2 : 10

2 What is 15 : 20 in simplest form? → Example 1.1

A. 5 : 4 **B.** 3 : 4
C. 1 : 5 **D.** 30 : 40

3 Adults : children = 5 : 9. There are 630 children.
How many adults?
Hint: 9 parts = 630. → Example 1.4

A. 70 **B.** 126 **C.** 280 **D.** 350

Short answer show your working in the box

1 Write each ratio in simplest form. *Hint: find the HCF. Decimals? Multiply by 10 first.* → Example 1.1

(a) 20 : 30 (b) 6 : 9 : 12 (c) 1.5 : 2.5

(a)	(b)	(c)

2 Consider the ratio 10 : 4. Write it → Examples 1.1 and 1.2

(a) in simplest form (b) in the form 1 : a (c) in the form b : 1

(a)	(b)	(c)

3 Find the value of the pronumeral. *Hint: divide to find what was multiplied.* → Example 1.3

(a) 2 : 3 = a : 15 (b) 4 : 3 = a : 24 (c) 3 : 5 = 12 : b

(a)	(b)	(c)

4 Gabriel mixes sand and cement in the ratio 5 : 2. He uses 4 buckets of cement. How many buckets of sand? *Hint: 2 parts = 4 buckets.* → Example 1.4

5 The ratio of Victoria's height to Willow's height is 8 : 7. Victoria is 176 cm tall. How tall is Willow? *Hint: 8 parts = 176 cm.* → Example 1.4

Extended response answer each part in order

1 A fruit punch recipe mixes orange juice and lemonade in the ratio 3 : 5. → Key idea and Example 1.4

(a) Write the ratio of **lemonade** to **orange juice**.

(b) Mia uses 600 mL of orange juice. How many millilitres is **one part**?

(c) How much lemonade does Mia need?

(d) How much punch does Mia make altogether? *Hint: add the orange juice and the lemonade.*

(a)	(b)	(c)	(d)

2 Capture–recapture

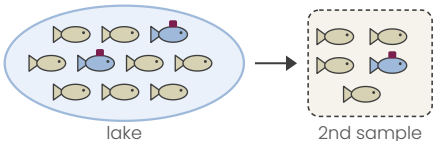
Key idea: Capture–recapture • Textbook 1B • MST-12-S2-05

Key idea Capture–recapture

Capture–recapture estimates the size of an animal **population** (how many animals there are) without counting them all.

1. **Capture** some animals. **Tag** them (put a mark or band on them) and let them go.
2. Wait for the tagged animals to **mix in** with the others.
3. **Recapture:** catch a 2nd **sample** (group). Count the tags.

The fraction with tags in the 2nd sample is about the same as the fraction with tags in the **whole population**.



lake 2nd sample

1 in 5 fish caught has a tag, so about 1 in 5 fish in the lake has a tag.

$$\frac{\text{tagged in 2nd sample}}{\text{size of 2nd sample}} = \frac{\text{tagged at first}}{\text{population}}$$

Assumptions

The estimate is only good if:

- no animals are **born**, **die**, **arrive** or **leave** between the two catches
- the **tags don't fall off**
- the tagged animals **mix in** with the others
- every animal has the **same chance** of being caught.

Finding the population

Is it 1 in something? If 1 in 5 have tags, the animals tagged at first are one fifth of the population. Multiply by 5.

Any numbers: use the rule.

$$\text{population} = \frac{\text{tagged at first} \times \text{size of 2nd sample}}{\text{tagged in 2nd sample}}$$

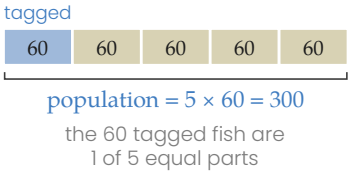
CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* use the capture–recapture technique to estimate a population. ○○○

WORKED EXAMPLE 2.1 Estimating a population

A ranger catches 60 fish in a lake, tags them and lets them go. A week later she catches 50 fish, and 10 of them have tags. Estimate the number of fish in the lake (a) using fractions (b) using the rule.

	WORKING	WHY
(a)	fraction tagged $= \frac{10}{50}$	10 of the 50 fish caught have tags.
	$= \frac{10 \div 10}{50 \div 10} = \frac{1}{5}$	Divide the top and bottom by **10**. **1 in 5** has a tag.
	60 fish $= \frac{1}{5}$ of all the fish	So about 1 in 5 of **all** the fish have tags.
	population $= 60 \times 5$	One fifth is 60, so all of it is **5 times** as big.
	$=$ 300 fish	Check: 60 out of 300 is 1 in 5.
(b)	population $= \frac{60 \times 50}{10} = \frac{3000}{10}$	Use the rule: 60 tagged, 50 caught, 10 with tags.
	$=$ 300 fish	3000 ÷ 10 = 300, the same as (a).



WORKED EXAMPLE 2.2 — Using the rule

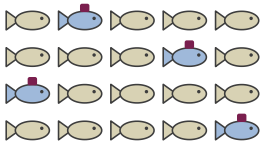
A scientist tags 30 frogs at a pond and lets them go. A week later she catches 25 frogs, and 10 of them have tags. (a) What fraction of the 2nd sample has tags? (b) Estimate the number of frogs at the pond.

WORKING	WHY
(a) fraction tagged $= \frac{10}{25} = \frac{10 \div 5}{25 \div 5}$	10 of the 25 frogs caught have tags. The HCF of 10 and 25 is **5**.
$= \boxed{\frac{2}{5}}$	Divide the top and bottom by 5. **2 in 5** have tags.
(b) population $= \frac{30 \times 25}{10} = \frac{750}{10}$	2 in 5 is not 1 in something, so use the **rule**. 30 × 25 = 750.
$= \boxed{75 \text{ frogs}}$	Divide: 750 ÷ 10 = 75. Check: 30 out of 75 is also 2 in 5 (÷ 15).

WORKED EXAMPLE 2.3 — Expected number tagged

About 500 fish live in a dam, and 100 of them have tags. Ali catches 20 fish. How many would you expect to have tags?

WORKING	WHY
$\frac{100}{500} = \frac{100 \div 100}{500 \div 100} = \frac{1}{5}$	100 of the 500 fish have tags. Divide the top and bottom by 100: **1 in 5** has a tag.
expected $= \frac{1}{5} \times 20 = 20 \div 5$	Expect the **same fraction** in Ali's catch. One fifth means ÷ 5.
$= \boxed{4 \text{ fish}}$	Check: 4 out of 20 is 1 in 5.



Ali's 20 fish: expect 1 in 5 to have a tag

WORKED EXAMPLE 2.4 — Why an estimate might be wrong

In Example 2.1, suppose some tags had fallen off, so only 6 of the 50 fish caught had tags, not 10. (a) Find the new estimate. (b) Is it too big or too small?

WORKING	WHY
(a) population $= \frac{60 \times 50}{6} = \frac{3000}{6}$	Use the rule with only **6** tagged. 60 × 50 = 3000.
$= \boxed{500 \text{ fish}}$	Divide: 3000 ÷ 6 = 500.
(b) 6 tagged, not 10	Lost tags mean **fewer** tagged fish are caught.
$\boxed{\text{The estimate is too big.}}$	Dividing by a smaller number gives a **bigger** answer: 500, not 300.

YOUR TURN 2

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Which would make the estimate **less** accurate?

→ Assumptions

A. The tagged fish mix in.
B. No fish are born or die.
C. Some tags fall off.
D. No fish leave the lake.

2 25 birds are tagged. Later, 40 birds are caught and 10 have tags. Estimate the population. → Example 2.1

A. 75 **B.** 100 **C.** 250 **D.** 1000

3 600 fish live in a lake. 60 have tags. Sam catches 30 fish. How many would you expect to have tags?
Hint: 1 in 10 has a tag.

→ Example 2.3

A. 3 **B.** 6 **C.** 10 **D.** 20

Short answer show your working in the box

1 Estimate each population. *Hint: find the fraction tagged in the 2nd sample first.* → Example 2.1

(a) 20 tagged at first. Later 30 caught, 10 tagged.

(b) 30 tagged at first. Later 20 caught, 5 tagged.

(c) 80 tagged at first. Later 60 caught, 6 tagged.

(a)

(b)

(c)

2 A biologist tags 50 carp in a dam and lets them go. Later he catches 40 carp, and 16 of them have tags. → Example 2.2

(a) What fraction of the 2nd sample has tags? Simplify it.

(b) Estimate the number of carp in the dam. *Hint: use the rule.*

(a)

(b)

3 A scientist tags 200 fish in a pond and lets them go. Later she catches 50 fish, and 20 of them have tags. Estimate the number of fish in the pond. *Hint: use the rule.* → Example 2.2

4 About 400 possums live in a forest, and 80 of them have tags. A ranger catches 30 possums. How many would you expect to have tags? *Hint: 80 out of 400 is 1 in 5.* → Example 2.3

5 Some of the tagged fish are eaten by birds before the 2nd catch. → Example 2.4

(a) Will **more** or **fewer** tagged fish be caught?

(b) Will the estimate be **too big** or **too small**?

(a)

(b)

Extended response answer each part in order

1 A ranger tags 40 kangaroos in a national park and lets them go. Two weeks later she catches 30 kangaroos, and 6 of them have tags. → Key idea, Assumptions and Example 2.1

(a) What fraction of the 2nd sample has tags? Simplify it.

(b) Estimate the number of kangaroos in the park. *Hint: use your answer to (a).*

(c) Why does the ranger wait two weeks before the 2nd catch?

(d) Write one other thing that must be true for her estimate to be good.

(a)

(b)

(c)

(d)

3 Dividing a quantity in a given ratio

Key idea: Dividing a quantity in a given ratio • Textbook 1C • MST-12-S2-05

Key idea Dividing a quantity in a ratio

To divide a **quantity** (an amount) in a ratio, change the ratio into **fractions**. Then multiply the quantity by each fraction.

1. **Add** the parts to get the total number of parts: $2 + 3 = 5$.
2. Write each part **over the total**: $\frac{2}{5}$ and $\frac{3}{5}$.
3. **Multiply** the quantity by each: $\frac{2}{5} \times \$40 = \$16$, $\frac{3}{5} \times \$40 = \$24$.
4. **Check** that the shares add up to the quantity: $\$16 + \$24 = \$40$.

Amy \$8 \$8 = \$16

Ben \$8 \$8 \$8 = \$24

Share \$40 in the ratio 2:3.
5 parts, so each part is \$8.

Ratio to fractions

Ratio	Total parts	Fractions
5 : 1	$5 + 1 = 6$	$\frac{5}{6}$ and $\frac{1}{6}$
1 : 3	$1 + 3 = 4$	$\frac{1}{4}$ and $\frac{3}{4}$
1 : 2 : 4	$1 + 2 + 4 = 7$	$\frac{1}{7}$, $\frac{2}{7}$ and $\frac{4}{7}$

Tips

- **Order matters.** Amy : Ben = 2 : 3 means Amy gets 2 parts and Ben gets 3.
- The **smaller** share goes with the **smaller** number.
- **Three parts?** Same steps. Add all three parts.

Another way: find one part

Divide the amount by the total parts to find **one part**. Then multiply by each number in the ratio.

$\$40 \div 5 = \mathbf{\$8} \rightarrow 2 \times \mathbf{\$8} = \$16,\ 3 \times \mathbf{\$8} = \$24$

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- *I can* divide a quantity in a given ratio, with two or three parts. ○○○

WORKED EXAMPLE 3.1 — Sharing in a ratio

Ava and Ben share \$45 in the ratio 4 : 5. (a) How much does Ava get? (b) How much does Ben get?

	WORKING	WHY
(a)	total parts $= 4 + 5 = 9$	**Add** the parts of the ratio.
	Ava $= \frac{4}{9} \times \$45$	Ava is first, so she gets **4** of the 9 parts.
	$= \$45 \div 9 \times 4$	Divide by 9, then multiply by 4.
	$= \$5 \times 4$	One part is $\$45 \div 9 = \$5$.
	$= \boxed{\$20}$	Ava gets 4 parts of \$5.
(b)	Ben $= \frac{5}{9} \times \$45$	Ben gets the other **5** parts.
	$= \$45 \div 9 \times 5$	Divide by 9, then multiply by 5.
	$= \$5 \times 5$	One part is still \$5.
	$= \boxed{\$25}$	Check: \$20 + \$25 = \$45, the whole amount.

Ava \$5 \$5 \$5 \$5 = \$20

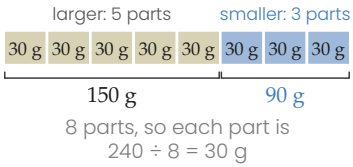
Ben \$5 \$5 \$5 \$5 \$5 = \$25

9 parts, so each part is
\$45 ÷ 9 = \$5

WORKED EXAMPLE 3.2 — Finding one share

A 240 g bag of nuts is divided in the ratio 5 : 3. What is the mass of the **smaller** amount?

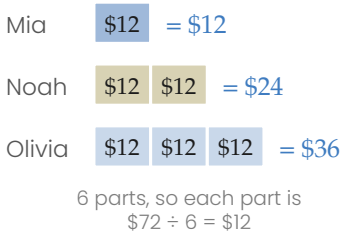
WORKING	WHY
total parts = 5 + 3 = 8	**Add** the parts.
smaller = $\frac{3}{8}$ × 240 g	The smaller amount goes with the **smaller** number, 3.
= 240 g ÷ 8 × 3 = 30 g × 3	Divide by 8, then multiply by 3.
= 90 g	Check: 5 × 30 = 150 g, and 150 + 90 = 240.



WORKED EXAMPLE 3.3 — A ratio with three parts

Mia, Noah and Olivia share $72 in the ratio 1 : 2 : 3. How much does each person get?

WORKING	WHY
total parts = 1 + 2 + 3 = 6	Add **all three** parts.
Mia = $\frac{1}{6}$ × $72 = $72 ÷ 6 = $12	Mia gets 1 part. One part is $12.
Noah = $\frac{2}{6}$ × $72 = $12 × 2 = $24	Noah gets 2 parts of $12.
Olivia = $\frac{3}{6}$ × $72 = $12 × 3 = $36	Olivia gets 3 parts of $12.
Mia $12, Noah $24, Olivia $36	Check: $12 + $24 + $36 = $72.



WORKED EXAMPLE 3.4 — Making the ratio first

A school camp has 45 students and 18 teachers. The 21 tents are shared in the ratio students : teachers. How many tents does each group get?

WORKING	WHY
students : teachers = 45 : 18	Write the numbers in the **same order** as the words.
= 45 ÷ 9 : 18 ÷ 9 = **5 : 2**	The HCF of 45 and 18 is **9**. Divide both parts by 9 to get simplest form.
total parts = 5 + 2 = 7	**Add** the parts.
1 part = 21 ÷ 7 = 3 tents	Share the 21 tents: find **one part**.
students = 5 × 3 = 15 tents	Students get 5 parts.
teachers = 2 × 3 = 6 tents	Teachers get 2 parts.
15 tents and 6 tents	Check: 15 + 6 = 21 tents.

YOUR TURN 3

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Kim and Lee share some money in the ratio 2 : 7. What fraction does Kim get?

→ Key idea

A. $\frac{2}{7}$ **B.** $\frac{2}{9}$ **C.** $\frac{7}{9}$ **D.** $\frac{1}{2}$

2 A 360 g bag of lollies is divided in the ratio 7 : 5. Find the mass of the smaller amount. → Example 3.2

A. 150 g **B.** 168 g **C.** 192 g **D.** 210 g

3 $90 is shared in the ratio 1 : 3 : 5. What is the largest share? *Hint: 1 + 3 + 5 = 9 parts.*

→ Example 3.3

A. $10 **B.** $18 **C.** $30 **D.** $50

Short answer show your working in the box

1 Divide \$400 in each ratio. *Hint: add the parts first.* → Example 3.1

(a) 3 : 2 (b) 4 : 1 (c) 3 : 7

(a) (b) (c)

2 Daniel and Eddie share their profits in the ratio 3 : 4. *Hint: 3 + 4 = 7 parts.* → Examples 3.1 and 3.2

(a) One week the profit is \$700. How much does Daniel get?

(b) The next week the profit is \$1400. How much does Eddie get?

(a) (b)

3 Share \$100 between Jack, Kate and Leo in the ratio 2 : 3 : 5. *Hint: 2 + 3 + 5 = 10 parts.* → Example 3.3

4 A 35 kg bag of potting mix has soil and compost in the ratio 4 : 3. How many kilograms of compost are in the bag? *Hint: 7 parts = 35 kg.* → Example 3.2

5 A 90 cm piece of wood is cut into two pieces in the ratio 7 : 2. How long is the longer piece? *Hint: 9 parts = 90 cm.* → Example 3.2

Extended response answer each part in order

1 A cruise ship has 160 passengers and 60 crew. Its 22 lifeboats are shared in the ratio passengers : crew. → Example 3.4

(a) Write the ratio passengers : crew in simplest form. *Hint: the HCF of 160 and 60 is 20.*

(b) How many parts are there in total?

(c) How many lifeboats is one part?

(d) How many lifeboats are for the passengers? How many are for the crew?

(a) (b) (c) (d)

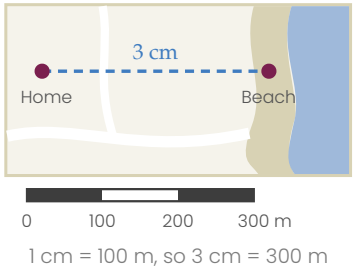
4 Scale drawings

Key idea: Scale drawing • Textbook 1D • MST-12-S2-05

Key idea Scale drawing

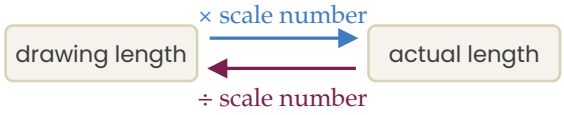
A **scale** compares a length on a drawing with the real (actual) length.

- **Scale = drawing length : actual length.** The drawing comes **first**.
- **With units:** 1 cm to 1 m (or 1 cm = 1 m). Each 1 cm on the drawing is 1 m in real life.
- **No units:** a ratio such as 1 : 100. Each 1 cm is 100 cm (or each 1 mm is 100 mm).
- 1 cm to 1 m is the **same scale** as 1 : 100, because 1 m = 100 cm.



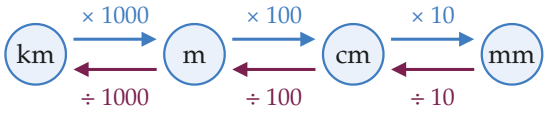
1 cm = 100 m, so 3 cm = 300 m

Using a scale



- The **scale number** is the second number: 50 in 1 : 50, and 20 in 1 cm = 20 m.
- Both lengths must be in the **same unit**. Change units first if you need to.

Changing units



To a **smaller** unit, **multiply**.
To a **bigger** unit, **divide**.
1 km = 1000 m = 100 000 cm

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* use a scale drawing to calculate real distances. ○○○
- ▸ *I can* use a scale to find a length on a drawing, and write a scale in the form 1 : x. ○○○

WORKED EXAMPLE 4.1 Finding a real length

Find each real length. (a) Scale 1 cm = 20 m, two trees 4.5 cm apart on a map (b) Scale 1 : 50, a wall 7 cm long on a plan, in metres (c) Scale 1 : 25 000, two lookouts 8 cm apart on a map, in kilometres

	WORKING	WHY
(a)	1 cm = 20 m	Every 1 cm on the map is **20 m** in real life.
	real = 4.5 × 20	**Multiply** the map length by 20.
	= 90 m	The scale says **m**, so the answer is in metres.
(b)	real = 7 × 50	No units: every 1 cm is **50 cm**. Multiply by 50.
	= 350 cm	The plan length is in cm, so this answer is in **cm**.
	= 350 ÷ 100	Change to metres: 100 cm = 1 m, so **divide** by 100.
	= 3.5 m	Check: 1 cm is 50 cm = 0.5 m, and 7 × 0.5 = 3.5 m.
(c)	real = 8 × 25 000	**Multiply** the map length by 25 000.
	= 200 000 cm	The map length is in cm, so this answer is in **cm**.
	= 200 000 ÷ 100 000	Change to km: 1 km = 100 000 cm, so **divide** by 100 000.
	= 2 km	Check: 4 cm × 25 000 = 100 000 cm = 1 km, so 8 cm = 2 km.

WORKED EXAMPLE 4.2 — Finding a length on the drawing

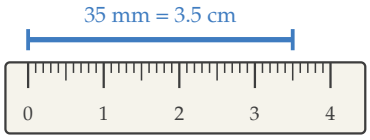
Find each length on the drawing. (a) Scale 1 cm = 25 m, a path 150 m long, on a map (b) Scale 1 : 40, a table 1.6 m long, on a plan, in centimetres

	WORKING	WHY
(a)	1 cm = 25 m	Every 25 m in real life is **1 cm** on the map.
	map = 150 ÷ 25	Real to map, so **divide** by 25.
	= 6 cm	The scale's first unit is cm. Check: 6 × 25 = 150 m.
(b)	1.6 m = 1.6 × 100 = 160 cm	Change to cm first, so both lengths use the **same unit**.
	plan = 160 ÷ 40	Real to plan, so **divide** by the scale number, 40.
	= 4 cm	Check: 4 × 40 = 160 cm = 1.6 m.

WORKED EXAMPLE 4.3 — Change the map length first

A map has the scale 1 cm = 2 km. Two towns are 35 mm apart on the map. How far apart are the towns in real life?

WORKING	WHY
35 mm = 35 ÷ 10	Change mm to cm: **divide** by 10.
= 3.5 cm	Now it is in cm, the same unit as the scale.
real = 3.5 × 2	Every 1 cm is 2 km, so **multiply** by 2.
= 7 km	The scale says **km**, so the answer is in kilometres.



ruler in cm: 10 mm = 1 cm

WORKED EXAMPLE 4.4 — Writing a scale as 1 : *x*

Write each scale in the form 1 : *x*. (a) 1 cm to 2 m (b) 1 mm to 5 cm (c) 1 m to 4 km

(a)	(b)	(c)	WHY
1 cm : 2 m	1 mm : 5 cm	1 m : 4 km	Write it as drawing : actual.
= 1 cm : 2 × 100 cm	= 1 mm : 5 × 10 mm	= 1 m : 4 × 1000 m	Make the units the **same**.
= 1 : 200	= 1 : 50	= 1 : 4000	Same units, so **drop** them.

YOUR TURN 4

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 A plan has the scale 1 : 100. What does 1 cm on the plan stand for? → Key idea

A. 1 cm **B.** 10 cm
C. 100 cm **D.** 100 m

2 A drawing has the scale 1 : 20. A line on it is 20 mm long. What is the real length?
→ Example 4.1

A. 1 mm **B.** 20 mm
C. 40 mm **D.** 400 mm

3 A map has the scale 1 mm = 150 m. Two places are 600 m apart. How far apart are they on the map?
Hint: divide by 150.
→ Example 4.2

A. 4 mm **B.** 0.25 mm
C. 4 cm **D.** 90 000 mm

Short answer show your working in the box

1 On a map, the scale is 1 cm = 50 m. How far apart in real life are two points that are this far apart on the map? *Hint: multiply by 50.* → Example 4.1

(a) 3.2 cm (b) 6 cm (c) 0.4 cm

(a)	(b)	(c)

2 A park map has the scale 1 : 400. Find each real distance in metres. *Hint: × 400, then ÷ 100.* → Example 4.1

(a) 5 cm (b) 20 cm (c) 1.5 cm

(a)	(b)	(c)

3 Write each scale in the form 1 : x. *Hint: change to the same unit first.* → Example 4.4

(a) 1 cm to 3 m (b) 1 mm to 6 cm (c) 1 m to 2.5 km

(a)	(b)	(c)

4 A kitchen plan has the scale 1 : 50. The bench is 3 m long. How long is the bench on the plan, in cm? *Hint: change 3 m to cm first.* → Example 4.2

5 A map has the scale 1 cm = 5 km. Two towns are 46 mm apart on the map. What is the real distance, in km?
Hint: 46 mm = 4.6 cm. → Example 4.3

Extended response answer each part in order

1 A bushwalking map has the scale 1 : 50 000. → Examples 4.1 and 4.2

(a) A walking track is 6 cm long on the map. How long is the real track, in centimetres?

(b) Change your answer to part (a) into kilometres. *Hint: 1 km = 100 000 cm.*

(c) A lake is 2.5 km long. Change 2.5 km into centimetres. *Hint: × 100 000.*

(d) How long is the lake on the map, in centimetres?

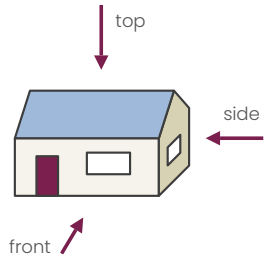
(a)	(b)	(c)	(d)

5 Plans and elevations

Key idea: Plans and elevations • Textbook 1E • MST-12-S2-05

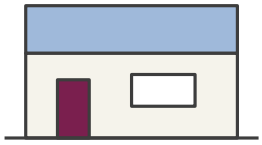
Key idea Plans and elevations

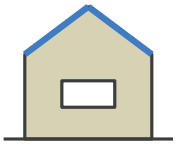
- A **plan** is the view of an object from the **top**: a bird's-eye view, looking straight down.
- An **elevation** is the view from **one side**, such as the **front elevation** or a **side elevation**.



the house


front
plan (from the top)


front elevation


side elevation

Building plans

- A **floor plan** is a plan of a building. It shows the walls, doors, windows and fittings.
- It is drawn to **scale**, usually 1 : 100: 1 mm on the plan is 100 mm in real life.
- Lengths on building plans are in **millimetres**. 1 cm = 10 mm, 1 m = 1000 mm.

At 1 : 100: 36 mm → 36 × 100 = 3600 mm = 3.6 m

Plan symbols


door

window

sliding door


robe (wardrobe)


bath


shower


toilet (W.C.)

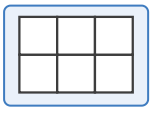
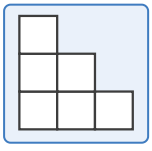
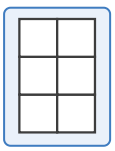

basin

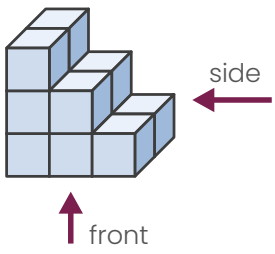
CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* tell a plan from an elevation, and read the symbols on a floor plan. ○○○
- ▸ *I can* use a scale on a building plan. ○○○

WORKED EXAMPLE 5.1 Drawing a plan and elevations

Some steps are made from 12 cubes. Draw their plan, front elevation and side elevation (from the right).

VIEW	DRAWING	WHY
Plan		Look straight **down** from the top. You see 3 squares across and 2 back.
Front elevation		Look straight at the **front**. The columns are 3, 2 and 1 cubes tall.
Side elevation		Look from the **right**. You see 2 squares across (the depth) and 3 up.

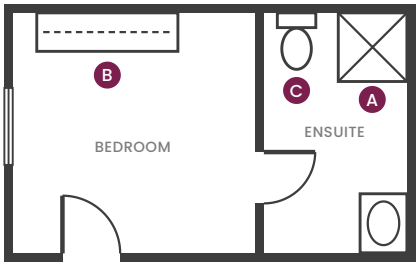


steps made from 12 cubes

WORKED EXAMPLE 5.2 — Reading a floor plan

This floor plan shows a bedroom and an ensuite (a small bathroom). Use the symbol key on page 16 to name the symbols labelled A, B and C.

	ANSWER	WHY
A	shower	A square with a cross is a **shower**.
B	robe (wardrobe)	A long box with a dashed line (the hanging rail) is a **robe**.
C	toilet (W.C.)	A small box (the cistern) and an oval bowl is a **toilet**. W.C. means water closet.



WORKED EXAMPLE 5.3 — Using the scale on a building plan

A house plan is drawn at 1 : 50. (a) What real distance, in cm, does 1 mm on the plan stand for? (b) A wall is 96 mm long on the plan. How long is the real wall, in metres?

	WORKING	WHY
(a)	real = 1 × 50 = 50 mm	1:50, so real lengths are **50 times** the plan.
	= 50 ÷ 10	Change mm to cm: 10 mm = 1 cm, so **divide** by 10.
	= 5 cm	Every 1 mm on the plan is 5 cm in real life.
(b)	real = 96 × 50	**Multiply** the plan length by the scale number, 50.
	= 4800 mm	The plan length is in mm, so this answer is in **mm**.
	= 4800 ÷ 1000	Change mm to m: 1000 mm = 1 m, so **divide** by 1000.
	= 4.8 m	Check: 96 × 5 cm = 480 cm = 4.8 m.

WORKED EXAMPLE 5.4 — Finding the scale of a plan

On a plan, a wall that is 4 m long is drawn 20 mm long. Find the scale of the plan, in the form 1 : x.

WORKING	WHY
20 mm : 4 m	Scale = drawing length : actual length.
= 20 mm : 4 × 1000 mm	Change to the **same unit**: 1 m = 1000 mm.
= 20 : 4000	Both parts are in mm, so drop the units.
= 20 ÷ 20 : 4000 ÷ 20	Divide **both** parts by the first number, 20.
= 1 : 200	Check: 20 mm × 200 = 4000 mm = 4 m.

YOUR TURN 5

Stuck? Each question shows you where to look.

Multiple choice — circle the best answer

1 A can of drink stands upright on a table. What shape is its **plan**? → Key idea

A. circle **B.** rectangle
C. square **D.** triangle

2 What does this floor plan symbol show?  → Example 5.2

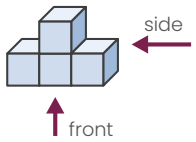
A. bath **B.** basin
C. shower **D.** toilet

3 On a plan drawn at 1 : 100, what real distance does 1 mm stand for? *Hint: 10 mm = 1 cm.* → Example 5.3

A. 1 cm **B.** 10 cm
C. 100 cm **D.** 1000 cm

Short answer show your working in the box

1 This object is made from 4 cubes. Sketch its plan, its front elevation and its side elevation (from the right). *Hint: the plan is the view from the top.* → Example 5.1



(a) plan (b) front elevation (c) side elevation

(a)	(b)	(c)

2 A house plan is drawn at 1 : 100. Find each real length, in metres. *Hint: × 100, then ÷ 1000.* → Example 5.3

(a) 30 mm on the plan (b) 45 mm on the plan (c) 62 mm on the plan

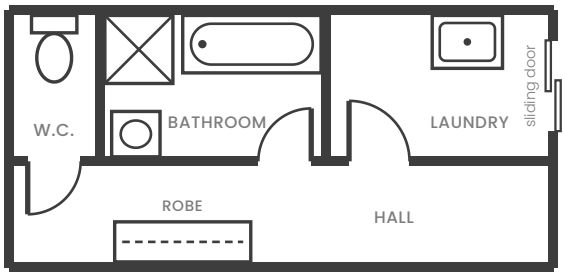
(a)	(b)	(c)

3 A plan is drawn at 1 : 100. A door is 9 mm wide on the plan. How wide is the real door, in cm? *Hint: 10 mm = 1 cm.* → Example 5.3

4 On a plan, a bed that is 2 m long is drawn 40 mm long. Find the scale of the plan, in the form 1 : x. *Hint: 2 m = 2000 mm.* → Example 5.4

Extended response answer each part in order

1 This floor plan is part of a house. On the plan, the laundry is 30 mm long and 20 mm wide. The real laundry is 3 m long. → Examples 5.2 to 5.4



(a) What does W.C. mean?

(b) Sketch the symbol used for the bath.

(c) Find the scale of the plan, in the form 1 : x. *Hint: 3 m = 3000 mm.*

(d) How wide is the real laundry, in metres?

(a)	(b)	(c)	(d)

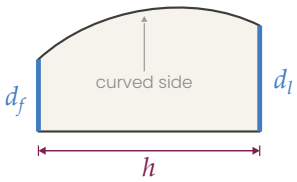
6 Perimeter and area of land

Key idea: Perimeter and area • Textbook 1F • MST-12-S2-05

Key idea Perimeter and area of land

Perimeter is the distance **around** the outside. **Area** is the space **inside**, in m^2.

- **Perimeter from a plan or an aerial photo** (a photo taken from above): multiply each side by the **scale**, then add.
- **Estimate an area:** cover it with a grid of squares and **count** them.
- **Calculate an area:** use a **formula** (rectangle, triangle, semicircle).
- **One curved side?** The **trapezoidal rule** gives an approximate area (≈).



a block with one curved side

Formulas you need

Shape	Area	Perimeter
Rectangle	length × width	add the 4 sides
Triangle	$\frac{1}{2}$ × base × height	add the 3 sides
Semicircle	$\pi r^2 \div 2$	curved edge = $\pi d \div 2$

Semicircle = **half** a circle. d = diameter, $r = d \div 2$

The trapezoidal rule

$$A \approx \frac{h}{2}(d_f + d_l)$$

- h = distance between the two measurements (the textbook calls it b)
- d_f = first measurement, d_l = last measurement
- **Two strips:** $A \approx \frac{h}{2}(d_1 + d_2) + \frac{h}{2}(d_2 + d_3)$

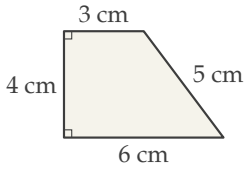
CHECKLIST · SUCCESS CRITERIA — Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* find the perimeter of land from a plan or an aerial photo, using the scale. ○○○
- ▸ *I can* find an area by counting squares, with a formula, or with the trapezoidal rule. ○○○

WORKED EXAMPLE 6.1 — Perimeter from a plan

This plan of a block of land has the scale 1 cm = 10 m. Find the real perimeter of the block.

WORKING	WHY
$6 \times 10 = 60$ m, $4 \times 10 = 40$ m	1 cm on the plan = 10 m, so **multiply** by 10.
$3 \times 10 = 30$ m, $5 \times 10 = 50$ m	Do the same for the other two sides.
$P = 60 + 40 + 30 + 50$	**Add** all four real sides.
= 180 m	Check: 6 + 4 + 3 + 5 = 18 cm, and 18 × 10 = 180.

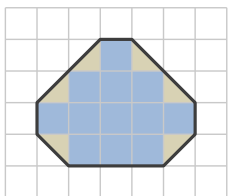


plan, scale 1 cm = 10 m

WORKED EXAMPLE 6.2 — Estimating area by counting squares

A paddock is drawn on a grid. Each square is 10 m by 10 m. Estimate the area of the paddock.

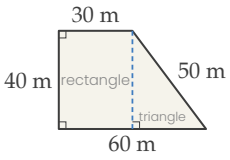
WORKING	WHY
12 full + 6 halves	Count **full** (blue) and **half** (sand) squares.
= 12 + 6 ÷ 2 = 15 squares	**Two halves** make one whole square.
1 square = 10 × 10 = 100 m^2	Find the area of **one** square.
$A \approx 15 \times 100$	Counting gives an **estimate**, so use ≈.
= 1500 m^2	


full half

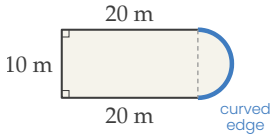
WORKED EXAMPLE 6.3 — Using the area formulas

Find (a) the area of the block in Example 6.1 (b) the perimeter of the lawn (c) the area of the lawn.

	WORKING	WHY
(a)	rectangle $= 30 \times 40 = 1200$ m^2	Split at the dashed line. **Length × width**.
	triangle base $= 60 - 30 = 30$ m	The rest of the 60 m bottom side.
	triangle $= \frac{1}{2} \times 30 \times 40 = 600$ m^2	$\frac{1}{2}$ **× base × height**.
	$A = 1200 + 600$	**Add** the two areas.
	$= \boxed{1800 \text{ m}^2}$	Area is in **square** metres.
(b)	curved edge $= \pi \times 10 \div 2 = 15.7$ m	**Half** of π × diameter. Use the π key.
	$P = 20 + 10 + 20 + 15.7$	Add the **outside** edges only.
	$= \boxed{65.7 \text{ m}}$	Do **not** add the dashed line.
(c)	rectangle $= 20 \times 10 = 200$ m^2	Rectangle: length × width.
	$r = 10 \div 2 = 5$ m	The radius is **half** the diameter.
	semicircle $= \pi \times 5^2 \div 2 = 39.3$ m^2	**Half** of πr^2. 39.26… rounds to 39.3.
	$A = 200 + 39.3$	**Add** the two parts.
	$= \boxed{239.3 \text{ m}^2}$	



the block, real lengths

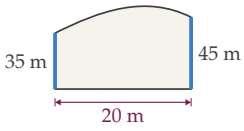


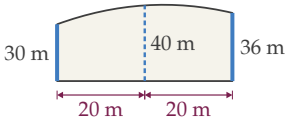
the lawn: a rectangle and a semicircle

WORKED EXAMPLE 6.4 — The trapezoidal rule

Use the trapezoidal rule to estimate the area of each block of land. (a) one strip (b) two strips

	WORKING	WHY
(a)	$A \approx \frac{20}{2}(35 + 45)$	$h = 20$, $d_f = 35$, $d_l = 45$.
	$= 10 \times 80$	Do 20 ÷ 2 and the **bracket** first.
	$= \boxed{800 \text{ m}^2}$	
(b)	$A \approx \frac{20}{2}(30 + 40) + \frac{20}{2}(40 + 36)$	Use the rule **twice**. 40 is in both.
	$= 10 \times 70 + 10 \times 76$	Do each ÷ 2 and each bracket.
	$= 700 + 760$	Multiply.
	$= \boxed{1460 \text{ m}^2}$	**Add** the two strips.





YOUR TURN 6

Stuck? Each question shows you where to look.

Multiple choice — circle the best answer

1 An aerial photo has the scale 1 cm = 20 m. A fence is 6 cm on the photo. How long is it really? → Example 6.1

A. 0.3 m **B.** 120 m
C. 26 m **D.** 1200 m

2 A paddock covers 8 full and 4 half squares. Each square is 100 m^2. What is its area? → Example 6.2

A. 800 m^2 **B.** 1200 m^2
C. 1000 m^2 **D.** 600 m^2

3 Use the trapezoidal rule: $h = 10$ m, $d_f = 18$ m, $d_l = 22$ m. *Hint: 10 ÷ 2 = 5.* → Example 6.4

A. 400 m^2 **B.** 45 m^2
C. 110 m^2 **D.** 200 m^2

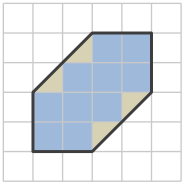
Short answer show your working in the box

1 An aerial photo of a rectangular block has the scale 1 cm = 20 m. On the photo the block is 4 cm long and 3 cm wide. Find the real lengths. *Hint: real length = photo length × 20.* → Examples 6.1 and 6.3

(a) Find the length and width. (b) Find the perimeter. (c) Find the area.

(a)	(b)	(c)

2 Each square on this grid is 10 m by 10 m. *Hint: two halves make one square.* → Example 6.2



(a) How many full squares?

(b) How many half squares?

(c) Estimate the area.

(a)	(b)	(c)

3 A garden bed is a semicircle with a diameter of 6 m. Find each one, to 1 decimal place. *Hint: the radius is 6 ÷ 2 = 3 m.* → Example 6.3

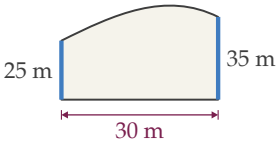
(a) the curved edge

(b) the perimeter (add the straight edge)

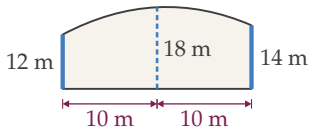
(c) the area

(a)	(b)	(c)

4 Use the trapezoidal rule to estimate the area of this block. → Example 6.4

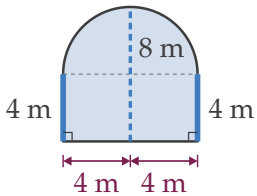


5 Use the trapezoidal rule **twice** to estimate the area of this block. → Example 6.4



Extended response answer each part in order

1 A concrete fish pond is a rectangle with a semicircle on one end. The rectangle is 8 m wide and 4 m long, so the semicircle has a diameter of 8 m. Give answers to 1 decimal place. → Examples 6.3 and 6.4



(a) Find the length of the curved edge. *Hint:* $\pi \times 8 \div 2$.

(b) Find the perimeter of the pond.

(c) Use the trapezoidal rule twice, with the measurements shown, to estimate the area of the pond. *Hint:* $A \approx \frac{4}{2}(4 + 8) + \frac{4}{2}(8 + 4)$.

(a)	(b)	(c)

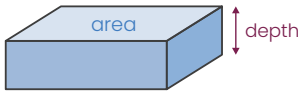
7 Volume of water

Key idea: Volume of water • Textbook 1F • MST-12-S2-05

Key idea Volume of water

When water has the **same depth** everywhere, its volume is the **area** of its surface times its **depth**.

- ***V* = area × depth.** The reference sheet writes $V = Ah$, where h is the depth.
- We write **depth** so it is not mixed up with the h in the trapezoidal rule.
- Find the **area** first: use a formula, or the trapezoidal rule for a curved side.
- Rain on an area works the same way: volume = area × rainfall depth.



water with the same depth everywhere

Rainfall

- Rain is measured in **millimetres**. Change it to **metres** first: **÷ 1000**.
- Volume of rain = area × rainfall depth, in m^3.
- Then change m^3 to litres: **× 1000**.

Underestimate or overestimate?

The trapezoidal rule uses **straight** lines. If the real edge curves **out**, the answer is too small (**underestimate**). If it curves **in**, the answer is too big (**overestimate**).

Changing units

Change	How	Example
$m^3 \to L$	× 1000	$2\ m^3 = 2 \times 1000 = 2000\ L$
$L \to m^3$	÷ 1000	$500\ L = 500 \div 1000 = 0.5\ m^3$
$mm \to m$	÷ 1000	$18\ mm = 18 \div 1000 = 0.018\ m$

$1\ m^3 = 1000\ L$ **$1\ m = 1000\ mm$** $m^2 \times m = m^3$

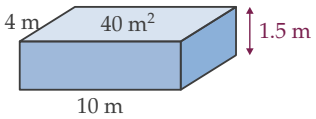
CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- *I can* find the volume of water from a plan, using the trapezoidal rule when necessary. ○○○
- *I can* find the volume of rain that falls on an area, in m^3 and in litres. ○○○
- *I can* say whether a trapezoidal rule estimate is an underestimate or an overestimate. ○○○

WORKED EXAMPLE 7.1 Volume = area × depth

A swimming pool is 10 m long and 4 m wide. The water is 1.5 m deep everywhere. Find the volume of water (a) in m^3 (b) in litres.

	WORKING	WHY
(a)	area $= 10 \times 4 = 40\ m^2$	The surface is a rectangle: **length × width**.
	V = area × depth	The depth is the **same** everywhere.
	$= 40 \times 1.5$	The depth is 1.5 m.
	$= \boxed{60\ m^3}$	$m^2 \times m$ gives **cubic** metres, m^3.
(b)	60×1000	$1\ m^3 = 1000\ L$, so **multiply** by 1000.
	$= \boxed{60\,000\ L}$	

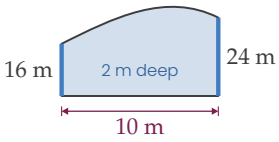


the water in the pool

WORKED EXAMPLE 7.2 — Trapezoidal rule, then volume

This dam has one curved side and is 2 m deep everywhere. Estimate (a) its area (b) its volume.

	WORKING	WHY
(a)	$A \approx \frac{10}{2}(16 + 24)$	Trapezoidal rule: $h = 10$, $d_f = 16$, $d_l = 24$.
	$= 5 \times 40$	Do 10 ÷ 2 and the **bracket** first.
	$=$ 200 m²	This is the area of the water's **surface**.
(b)	V = area × depth	The depth is 2 m. It is **not** $h = 10$.
	$= 200 \times 2$	Use the area from part (a).
	$=$ 400 m³	m² × m gives m³.

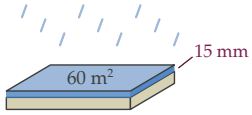


plan of the dam (not to scale)

WORKED EXAMPLE 7.3 — Volume of rain

15 mm of rain falls on a shed roof of area 60 m². Find the volume of rain (a) in m³ (b) in litres.

	WORKING	WHY
(a)	15 mm = 15 ÷ 1000 = 0.015 m	Change mm to m: **divide** by 1000.
	$V = 60 \times 0.015$	Area × rainfall depth. Both are in **metres**.
	$=$ 0.9 m³	Rain is a very **thin** layer of water.
(b)	0.9×1000	1 m³ = 1000 L, so **multiply** by 1000.
	$=$ 900 L	A tank under the roof could collect 900 L.

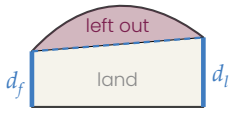


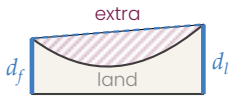
15 mm of rain on the roof

WORKED EXAMPLE 7.4 — Underestimate or overestimate?

The trapezoidal rule is used for each block. Is the estimate an underestimate or an overestimate?

	WORKING	WHY
(a)	rule uses the dashed line	The rule joins the ends with a **straight** line.
	the curve is **outside** it	So the rule leaves out the shaded land.
	underestimate	Too small. A volume from it is too small too.
(b)	the curve is **inside** it	So the rule adds the hatched part: not land.
	overestimate	Too big. A volume from it is too big too.





YOUR TURN 7

Stuck? Each question shows you where to look.

Multiple choice — circle the best answer

1 How many litres are in 3 m³?
→ Changing units

A. 30 L **B.** 300 L
C. 3000 L **D.** 0.003 L

2 What is 25 mm in metres?
→ Example 7.3

A. 2.5 m **B.** 0.025 m
C. 0.25 m **D.** 25 000 m

3 A pond has a surface area of 40 m² and is 0.5 m deep. Find its volume.
Hint: area × depth. → Example 7.1

A. 80 m³ **B.** 40.5 m³
C. 20 000 m³ **D.** 20 m³

Short answer show your working in the box

1 Change each one to the unit shown. *Hint: × 1000 or ÷ 1000.* → Changing units

(a) 5 m^3 to litres (b) 2500 L to m^3 (c) 40 mm to metres

(a)	(b)	(c)

2 A pool is 8 m long and 5 m wide. The water is 2 m deep everywhere. *Hint: area = 8 × 5.* → Example 7.1

(a) Find the area of the surface. (b) Find the volume of water. (c) Change the volume to litres.

(a)	(b)	(c)

3 A body of water has one curved side. Its measurements are $h = 6$ m (the textbook writes $b = 6$ m), $d_f = 20$ m and $d_l = 40$ m. The water is 2.5 m deep everywhere. *Hint: the depth is 2.5 m, not h.* → Example 7.2

(a) Use the trapezoidal rule to find the area. (b) Find the volume of water, in m^3.

(a)	(b)

4 Now 9 mm of rain falls on the same area as in Question 3. → Example 7.3

(a) Change 9 mm to metres. (b) Find the volume in m^3. (c) Change it to litres.

(a)	(b)	(c)

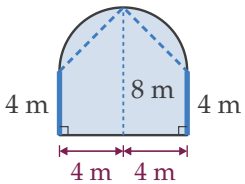
Extended response answer each part in order

1 A concrete fish pond is a rectangle with a semicircle on one end. Using the trapezoidal rule, its area is about 48 m^2. The water is 1.5 m deep everywhere. → Examples 7.1 and 7.4

(a) Find the volume of water in the pond, in m^3.

(b) How many litres of water are needed to fill the pond? *Hint: 1 m^3 = 1000 L.*

(c) Is your answer to (b) an underestimate or an overestimate? Explain. *Hint: compare the dashed lines with the curved edge.*



the dashed lines are the ones the trapezoidal rule uses

(a)	(b)	(c)

8 Rates

Key idea: Rate • Textbook 1G • MST-12-S2-05

Key idea Rate

A **rate** compares two amounts with **different units**, such as km/h, $/kg, L/100 km and $/hour.



- Say the / as "**per**", which means "for each". 60 km/h is 60 kilometres per hour: 60 km in each hour.
- **Rate or ratio?** A ratio compares amounts of the **same kind** and has no units. A rate has two **different** units, so **always write the units**.
- **Simplest form:** give the rate for **1** of the second unit. $12 for 3 kg = $4 for 1 kg = **$4/kg**.

Units you need

Measure	Units	
Time	1 h = 60 min	1 min = 60 s
Volume	1 L = 1000 mL	
Mass	1 kg = 1000 g	
Length	1 km = 1000 m	1 cm = 10 mm

Converting a rate

Change **one unit at a time**, and write every step.

- **Before the /**: a **smaller** unit? Multiply. A **bigger** unit? Divide. mL → L is ÷ 1000.
- **After the /**: ask "how much in **1** of the new unit?" 1 min = 60 s, so per s → per min is × 60.

Two units to change

Change the time first. Then change the other unit.

90 m/min = 90 × 60 m/h = 5400 m/h = 5400 ÷ 1000 km/h = 5.4 km/h

CHECKLIST · SUCCESS CRITERIA — Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* write a rate in simplest form, such as $4/kg. ○○○
- ▸ *I can* convert a rate to other units of measurement. ○○○

WORKED EXAMPLE 8.1 — Simplest form of a rate

Write each rate in simplest form. (a) $12 for 3 kg (b) 20 L in 4 min (c) 150 km in 2 h

	WORKING	WHY
(a)	$12 for 3 kg	We want the cost of **1** kg, not 3 kg.
	= $12 ÷ 3 for 1 kg	**Divide** by 3 to go down to 1 kg.
	= $4/kg	"For 1 kg" is written **/kg**. Say "4 dollars per kilogram".
(b)	20 L in 4 min	We want the litres in **1** minute.
	= 20 ÷ 4 L in 1 min	**Divide** by 4 to go down to 1 minute.
	= 5 L/min	Say "5 litres per minute".
(c)	150 km in 2 h	We want the distance in **1** hour.
	= 150 ÷ 2 km in 1 h	**Divide** by 2 to go down to 1 hour.
	= 75 km/h	Keep the **units**: 75 on its own does not say what it measures.

WORKED EXAMPLE 8.2 — Changing the time unit

Convert each rate. (a) 4 L/s to L/min (b) 4800 m/h to m/min

	WORKING	WHY
(a)	4 L/s	4 litres in **each** second.
	$= 4 \times 60$ L/min	There are **60** seconds in 1 minute, so **× 60**.
	= 240 L/min	A minute is longer, so the number gets **bigger**.
(b)	4800 m/h	4800 m in each hour, and 1 hour = 60 minutes.
	$= 4800 \div 60$ m/min	A minute is 60 times **shorter** than an hour, so **÷ 60**.
	= 80 m/min	Check: 80 × 60 = 4800 m in an hour.

WORKED EXAMPLE 8.3 — Changing the other unit

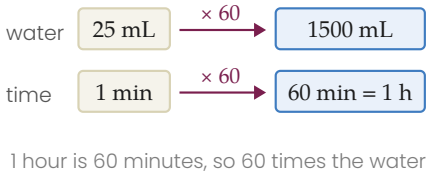
Convert each rate. (a) 4 cm/s to mm/s (b) 5000 m/h to km/h (c) \$20/kg to \$/g

	WORKING	WHY
(a)	4 cm/s $= 4 \times 10$ mm/s	1 cm = 10 mm. To a **smaller** unit, so **× 10**.
	= 40 mm/s	Only the cm changed. The /s stays the same.
(b)	5000 m/h $= 5000 \div 1000$ km/h	1000 m = 1 km. To a **bigger** unit, so **÷ 1000**.
	= 5 km/h	The /h stays the same.
(c)	\$20/kg = \$20 for 1000 g	This time the unit **after** the / changes. 1 kg = 1000 g.
	= \$20 ÷ 1000 for 1 g	**Divide** by 1000 to find the cost of **1** g.
	= \$0.02/g	1 g is much smaller than 1 kg, so it costs much **less**.

WORKED EXAMPLE 8.4 — Changing both units

A dripping tap wastes 25 mL of water every minute. Convert this rate to litres per hour (L/h).

WORKING	WHY
25 mL/min	**One unit at a time.** Start with the time.
$= 25 \times 60$ mL/h	There are 60 minutes in 1 hour, so **× 60**.
= 1500 mL/h	Now change mL to L.
$= 1500 \div 1000$ L/h	1000 mL = 1 L. To a **bigger** unit, so **÷ 1000**.
= 1.5 L/h	The tap wastes 1.5 litres every hour.



1 hour is 60 minutes, so 60 times the water

YOUR TURN 8

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Ella is paid \$96 for 4 hours of work. What is her rate of pay?
→ Example 8.1

A. \$24/h **B.** \$92/h
C. \$100/h **D.** \$384/h

2 A hose pours out 30 L of water every minute. What is this rate in L/h? → Example 8.4

A. 0.5 L/h **B.** 90 L/h
C. 1800 L/h **D.** 3000 L/h

3 Which is the **slowest** speed?
Hint: change each to km/h.
→ Examples 8.3 and 8.4

A. 40 km/h **B.** 6000 m/h
C. 60 km/h **D.** 2000 m/min

Short answer show your working in the box

1 Write each rate in simplest form. *Hint: divide to go down to 1 of the second unit.* → Example 8.1

(a) \$18 for 3 kg (b) 35 L in 5 min (c) 240 km in 4 h

(a)	(b)	(c)

2 Convert each rate. *Hint: 1 min = 60 s and 1 h = 60 min.* → Examples 8.2 and 8.4

(a) 5 bottles/s to bottles/min (b) 14 400 m/h to m/min (c) 50 mL/min to mL/h

(a)	(b)	(c)

3 Convert each rate. *Hint: 1 cm = 10 mm, 1 km = 1000 m and 1 kg = 1000 g.* → Example 8.3

(a) 8 cm/s to mm/s (b) 7000 m/h to km/h (c) \$15/kg to \$/g

(a)	(b)	(c)

4 A dripping tap wastes 30 mL of water every minute. Convert this rate to litres per hour (L/h). *Hint: × 60 first, then ÷ 1000.* → Example 8.4

5 A toy car travels at 120 cm/s. Convert this speed to mm/min. *Hint: change the time first (× 60), then cm to mm (× 10).* → Examples 8.2 and 8.3

Extended response answer each part in order

1 A machine in a factory fills 1200 bottles in 4 minutes. → Examples 8.1, 8.2 and 8.4

(a) Write this rate in simplest form, in bottles/min.

(b) Convert your answer to (a) into bottles/h.

(c) Convert your answer to (a) into bottles/s. *Hint: a second is 60 times shorter than a minute.*

(d) A new machine fills 6 bottles/s. Which machine is faster? *Hint: compare with (c).*

(a)	(b)	(c)	(d)

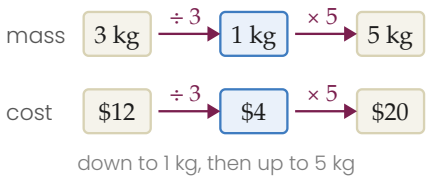
9 The unitary method and best buys

Key idea: Unitary method • Textbook 1G • MST-12-S2-05

Key idea Unitary method

The **unitary method** uses a rate to find an amount. *Unit* means **one**: go down to 1, then up to the amount you need.

1. **Divide** to find the amount for **1** (1 kg, 1 person, 1 hour).
   3 kg costs \$12, so 1 kg: \$12 ÷ 3 = \$4.
2. **Multiply** to find the amount you need. 5 kg: 5 × \$4 = \$20.



Setting out

Step	Working
Write what you know	3 kg costs \$12
Divide to go down to 1	1 kg: \$12 ÷ 3 = \$4
Multiply to go up	5 kg: 5 × \$4 = \$20

- Put the amount you want to find on the **right**.
- **Check:** more kilograms should cost more.

Best buys

- The **unit price** is the price for 1 unit, such as per 1 L, per 1 kg or per 100 g.
- Unit price = price ÷ amount
- Find the unit price of **each** item. The **lowest** unit price is the **best buy** (the best value).
- Supermarket shelf labels show the unit price under the price.

Price per 100 g

Count the lots of 100 g. Then divide the price by the number of lots.

500 g for \$4.50 → 500 ÷ 100 = 5 lots → \$4.50 ÷ 5 = **\$0.90 per 100 g**

CHECKLIST · SUCCESS CRITERIA — Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* use the unitary method to solve problems involving rates. ○○○
- ▸ *I can* work out which product is the best buy. ○○○

WORKED EXAMPLE 9.1 — The unitary method

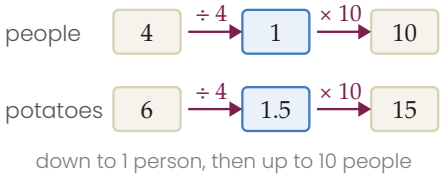
(a) 6 m of rope costs \$27. Find the cost of 10 m.
(b) A printer prints 90 pages in 5 minutes. How many pages does it print in 12 minutes?

	WORKING	WHY
(a)	6 m costs \$27	Write what you know. The cost is what we want, so it goes on the **right**.
	1 m: \$27 ÷ 6 = \$4.50	**Divide** by 6 to go down to **1** m.
	10 m: 10 × \$4.50	**Multiply** by 10 to go up to 10 m.
	= \$45	Check: 10 m is more than 6 m, so it costs more than \$27.
(b)	5 min: 90 pages	The pages are what we want, so they go on the right.
	1 min: 90 ÷ 5 = 18 pages	**Divide** by 5 to go down to **1** minute.
	12 min: 12 × 18	**Multiply** by 12 to go up to 12 minutes.
	= 216 pages	Check: more time, so more than 90 pages.

WORKED EXAMPLE 9.2 — When the amount for 1 is a decimal

A recipe for 4 people uses 6 potatoes. How many potatoes are needed for 10 people?

WORKING	WHY
4 people: 6 potatoes	Put what we want on the **right**.
1 person: 6 ÷ 4 = 1.5	**Divide** by 4. Half a potato is fine as a step.
10 people: 10 × 1.5	**Multiply** by 10. Do not round the 1.5.
= 15 potatoes	Check: more people need more potatoes.



down to 1 person, then up to 10 people

WORKED EXAMPLE 9.3 — Finding a time

A baker ices 1 cake every 15 minutes. How many hours does it take to ice 20 cakes?

WORKING	WHY
1 cake: 15 min	The time for **1** cake is given. No need to divide.
20 cakes: 20 × 15 min	**Multiply** by 20 to go up to 20 cakes.
= 300 min	The question asks for **hours**.
= 300 ÷ 60 h	60 min = 1 h, so **÷ 60**.
= 5 h	Check: 4 cakes an hour, and 4 × 5 = 20 cakes.



1 h = 60 min

4 cakes in each hour

WORKED EXAMPLE 9.4 — Best buys

Which is the best buy? (a) 2 L for $5.00 or 3 L for $6.60 (b) 400 g for $3.20 or 600 g for $4.20

	WORKING	WHY
(a)	2 L: $5.00 ÷ 2 = $2.50 per L	Find the price of **1** L: price ÷ litres.
	3 L: $6.60 ÷ 3 = $2.20 per L	Do the same for the other size.
	3 L for $6.60 is the best buy	$2.20 per L is the **lowest** unit price.
(b)	400 g: 400 ÷ 100 = 4 lots	Count the lots of **100 g** in the small size.
	$3.20 ÷ 4 = $0.80 per 100 g	Divide the price by the number of lots.
	600 g: 600 ÷ 100 = 6 lots	Count the lots of 100 g in the big size.
	$4.20 ÷ 6 = $0.70 per 100 g	Divide the price by the number of lots.
	600 g for $4.20 is the best buy	$0.70 per 100 g is the **lowest** unit price.

YOUR TURN 9

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 4 kg of apples cost $10. How much do 6 kg of apples cost?
→ Example 9.1

A. $2.50 **B.** $12
C. $15 **D.** $60

2 Which bottle of milk is the best buy? → Example 9.4

A. 1 L: $2.50 **B.** 2 L: $4.60
C. 3 L: $7.20 **D.** 4 L: $8.40

3 A courier delivers 1 parcel every 25 minutes. How many hours for 24 parcels?
Hint: 24 × 25 min, then ÷ 60.
→ Example 9.3

A. 10 **B.** 11 **C.** 12 **D.** 13

Short answer show your working in the box

1 A 5 kg bag of rice costs $9.20. Find the cost of each amount. *Hint: divide for (a), then multiply.* → Example 9.1

(a) 1 kg (b) 3 kg (c) 10 kg

(a)	(b)	(c)

2 20 m of curtain material costs $580. *Hint: go down to 1 m, then up to 35 m.* → Example 9.1

(a) What is the cost of 1 m? (b) What is the cost of 35 m?

(a)	(b)

3 A cake for 6 people needs 3 eggs. How many eggs are needed for a cake for 14 people? *Hint: 1 person: 3 ÷ 6 eggs.* → Example 9.2

4 Jess types 1 page every 12 minutes. How many hours does it take her to type 15 pages? → Example 9.3

5 A drink comes in two sizes: 1.25 L for $3.25 or 2 L for $4.20. → Example 9.4

(a) Find the price per litre of the 1.25 L bottle. (b) Find the price per litre of the 2 L bottle. (c) Which is the best buy?

(a)	(b)	(c)

Extended response answer each part in order

1 Instant coffee comes in a 200 g jar for $9.00 or a 500 g jar for $20.00. → Key idea and Example 9.4

(a) How many lots of 100 g are in each jar?

(b) Find the price per 100 g of the 200 g jar.

(c) Find the price per 100 g of the 500 g jar.

(d) Which jar is the best buy? *Hint: look for the lowest price per 100 g.*

(a)	(b)	(c)	(d)

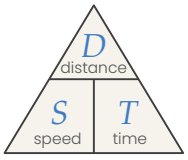
10 Speed and distance–time graphs

Key idea: Speed • Textbook 1G and 1H • MST-12-S2-05

Key idea Speed

Speed is a rate: 60 km/h means 60 km travelled in **each** hour.

- **Speed** = distance ÷ time: $S = \frac{D}{T}$. Turn it around: $D = S \times T$ and $T = \frac{D}{S}$.
- **Units:** km and h give **km/h**; m and s give **m/s**. Always write the units.
- On a **distance–time graph**, a **steeper** line is **faster** and a **flat** line means **stopped**.



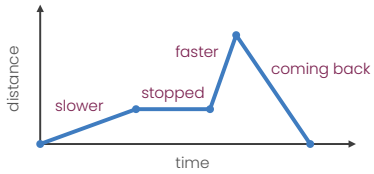
cover the one you want

Changing units

Change	How	Example
minutes → hours	**÷ 60**	45 min = 45 ÷ 60 = 0.75 h
hours → minutes	**× 60**	0.5 h = 0.5 × 60 = 30 min
km/h → m/s	**÷ 3.6**	18 km/h = 18 ÷ 3.6 = 5 m/s

- 15 min = 0.25 h, 30 min = 0.5 h, 45 min = 0.75 h
- **Why 3.6?** 1 m/s is 3600 m in an hour, or 3.6 km/h.
- **Comparing?** Use the **same units**, or compare the **times** for the same distance.

Distance–time graphs



- A line going **down** means coming **back** home.
- For **each part**, speed = distance ÷ time.
- **Average speed** = total distance ÷ total time. **Include** any stops.

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* solve problems involving speed, distance and time. ○○○
- ▸ *I can* read a distance–time graph and use it to find speeds. ○○○

WORKED EXAMPLE 10.1 Speed, distance and time

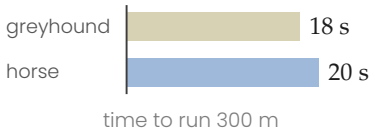
(a) A bus travels 36 km in 45 min. Find its speed in km/h.
(b) A train travels at 80 km/h for 2.5 h. How far does it go?
(c) Ana walks 10 km at 4 km/h. How long does she take? Give the time in hours and minutes.

	WORKING	WHY
(a)	$T = 45 \div 60 = 0.75$ h	For km/h, the time must be in **hours**. Minutes **÷ 60**.
	$S = \frac{D}{T} = 36 \div 0.75$	We want the **speed**. Cover S in the triangle: D is over T.
	$=$ 48 km/h	km and h give **km/h**. Check: 48 × 0.75 = 36 km.
(b)	$D = S \times T$	We want the **distance**. Cover D: S and T are side by side, so **multiply**.
	$= 80 \times 2.5$	The time is already in hours.
	$=$ 200 km	Check: 2 h is 160 km, and half an hour more is 40 km.
(c)	$T = \frac{D}{S} = 10 \div 4 = 2.5$ h	We want the **time**. Cover T: D is over S.
	0.5 h $= 0.5 \times 60 = 30$ min	Change the part of an hour to minutes: **× 60**.
	$T =$ 2 h 30 min	2.5 h is **not** 2 h 50 min.

WORKED EXAMPLE 10.2 — km/h and m/s

A horse gallops at 54 km/h. A greyhound runs 300 m in 18 s. Which would win a 300 m race?

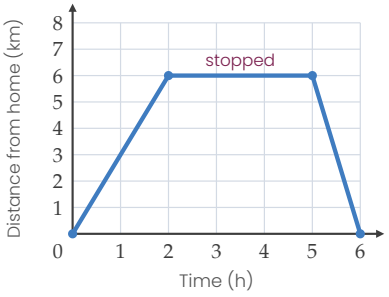
WORKING	WHY
54 km/h = 54 ÷ 3.6 m/s	The race is in **metres**, so change to **m/s**.
= 15 m/s	The horse runs 15 m every second.
horse: $T = 300 \div 15 = 20$ s	$T = D \div S$. m ÷ m/s gives **seconds**.
The greyhound wins.	It takes 18 s. The **shorter** time wins.



WORKED EXAMPLE 10.3 — Reading a distance–time graph

Jo walks from home to a friend's house and back. Use the graph to find (a) Jo's speed on the way there (b) how long she stopped (c) her speed on the way home.

	WORKING	WHY
(a)	$S = 6 \div 2$	The line goes **up** 6 km in the first 2 h.
	= 3 km/h	Speed = distance ÷ time, for that part.
(b)	stop = 5 − 2	The line is **flat** from 2 h to 5 h.
	= 3 h	A flat line means **stopped**.
(c)	$S = 6 \div 1$	It comes **down** 6 km, from 5 h to 6 h.
	= 6 km/h	A **steeper** line, so faster.



Jo's walk

WORKED EXAMPLE 10.4 — Average speed

Use Jo's walk in Example 10.3. Find her average speed (a) for the whole trip (b) without the stop.

	WORKING	WHY
(a)	$D = 6 + 6 = 12$ km	Total distance: there **and** back.
	$S = 12 \div 6$	The whole trip takes 6 h, **including** the stop.
	= 2 km/h	Average speed = total distance ÷ total time.
(b)	$T = 6 - 3 = 3$ h	Take away the 3 h stop.
	$S = 12 \div 3$	The same 12 km, in less time.
	= 4 km/h	Faster than (a), because the stop is left out.

YOUR TURN 10

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 A car travels 210 km in 3 h. What is its average speed?

→ Example 10.1

A. 0.014 km/h **B.** 70 km/h
C. 207 km/h **D.** 630 km/h

2 Which distance–time graph shows a car that is **stopped**?

→ Key idea

A.  **B.** 
C.  **D.** 

3 Leah runs 5 km in 30 minutes. What is her speed in km/h?
Hint: 30 min = 0.5 h.

→ Example 10.1

A. 0.17 km/h **B.** 2.5 km/h
C. 10 km/h **D.** 150 km/h

Short answer show your working in the box

1 Find the speed, distance or time. *Hint: cover the one you want in the triangle.* → Example 10.1

(a) A cyclist rides 42 km in 3 h. Find her speed.

(b) A plane flies at 650 km/h for 2 h. How far does it fly?

(c) A ship sails 150 km at 30 km/h. How long does it take?

(a)

(b)

(c)

2 Find each answer. *Hint: minutes ÷ 60 = hours, and hours × 60 = minutes.* → Example 10.1

(a) Ben cycles 4 km in 15 min. Find his speed in km/h.

(b) A ferry travels 21 km in 45 min. Find its speed in km/h.

(c) Zoe drives 135 km at 60 km/h. Find the time in hours and minutes.

(a)

(b)

(c)

3 A motorbike travels at 100 km/h. How far does it travel in 3.5 hours?
Hint: $D = S \times T$. → Example 10.1

4 Change each speed to m/s. → Example 10.2

(a) 90 km/h

(b) 72 km/h

(a)

(b)

5 Cathy Freeman's best time for the 400 m race was 48.63 s. A wombat can run at 36 km/h. → Example 10.2

(a) Change 36 km/h to m/s.

(b) How long would the wombat take to run 400 m?

(c) Would the wombat beat Cathy Freeman?

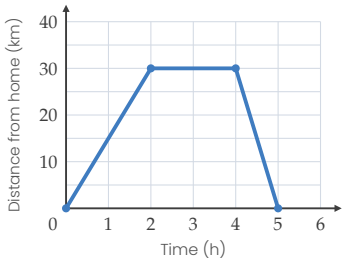
(a)

(b)

(c)

Extended response answer each part in order

1 Mia rides her bike from home to the beach, has a rest, then rides home. The graph shows her trip. → Examples 10.3 and 10.4



(a) What was her speed on the way to the beach?

(b) What was her speed on the way home?

(c) What was her average speed for the whole trip, including the rest?
Hint: total distance = 30 + 30 km.

(d) What was her average speed without the rest?
Hint: take away the rest time.

(a)

(b)

(c)

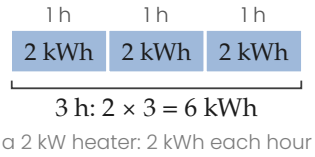
(d)

11 Energy

Key idea: Energy rate • Textbook 11 • MST-12-S2-05

Key idea Energy rate

Power is how fast an appliance uses electricity, in **watts** (W) or **kilowatts** (kW). **Energy** is the electricity it uses over time, in **kilowatt-hours** (kWh).



- **1 kW = 1000 W.** To change W to kW, **÷ 1000**.
  600 W = 600 ÷ 1000 = 0.6 kW
- **Energy (kWh) = power (kW) × time (h).** A 2 kW heater for 3 h uses 2 × 3 = 6 kWh.
- **Running cost = energy (kWh) × price per kWh.** At $0.30 per kWh: 6 × $0.30 = $1.80

Power ratings

Appliance	Power (W)	Power (kW)
Phone charger	10 W	0.01 kW
TV	150 W	0.15 kW
Microwave	1000 W	1 kW
Kettle	2200 W	2.2 kW

Watch the units

- **Wh or kWh?** W × h gives **watt-hours** (Wh). kW × h gives **kilowatt-hours** (kWh). 1 kWh = 1000 Wh.
- **Minutes?** Change to hours: **÷ 60**. 30 min = 0.5 h.
- **Cents?** Change to dollars: **÷ 100**. 30 c = $0.30.

Working backwards

Cost ÷ price per kWh = **energy** (kWh).
Energy (kWh) ÷ power (kW) = **time** (h).

$1.80 ÷ $0.30 = 6 kWh → 6 kWh ÷ 2 kW = **3 h**

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* calculate the energy used by an appliance, in kWh or Wh. ○○○
- ▸ *I can* calculate the cost of running electrical appliances. ○○○

WORKED EXAMPLE 11.1 Energy in kilowatt-hours

How much energy does each appliance use, in kWh?
(a) a 3.5 kW air conditioner for 2 h (b) a 1500 W heater for 3 h (c) a 40 W light for 10 h

	WORKING	WHY
(a)	Energy = 3.5 × 2 = 7 kWh	The power is already in **kW**. Energy = power × time. kW × h gives **kilowatt-hours** (kWh).
(b)	Power = 1500 ÷ 1000 = 1.5 kW Energy = 1.5 × 3 = 4.5 kWh	The power is in **watts**. Change to kW: **÷ 1000**. Energy = power (kW) × time (h). Check: 1.5 kWh each hour, for 3 hours.
(c)	Power = 40 ÷ 1000 = 0.04 kW Energy = 0.04 × 10 = 0.4 kWh	Change to kW: **÷ 1000**. **Careful:** 0.04, not 0.4. Energy = power (kW) × time (h). A light uses much less energy than a heater.

WORKED EXAMPLE 11.2 — Energy in watt-hours

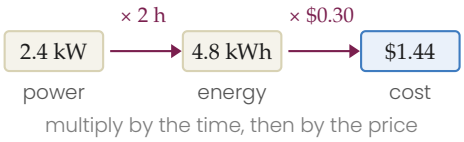
(a) A 60 W fan runs for 8 h. Find the energy it uses in Wh. (b) Write your answer to (a) in kWh.
(c) A 2000 W oven is used for 30 minutes. Find the energy it uses in Wh.

	WORKING	WHY
(a)	Energy = 60 × 8	The power is in **W** and we want **Wh**, so keep the watts.
	= 480 Wh	W × h gives **watt-hours** (Wh).
(b)	Energy = 480 ÷ 1000	1 kWh = 1000 Wh. To a **bigger** unit, so **÷ 1000**.
	= 0.48 kWh	Check: 0.06 kW × 8 h = 0.48 kWh.
(c)	Time = 30 ÷ 60 = 0.5 h	Time must be in **hours**. 1 h = 60 min, so **÷ 60**.
	Energy = 2000 × 0.5	Power in **W** × time in **h**.
	= 1000 Wh	That is the same as 1 kWh.

WORKED EXAMPLE 11.3 — Running cost

A 2.4 kW clothes dryer runs for 2 h. Electricity costs 30 c per kWh. Find the cost.

WORKING	WHY
Energy = 2.4 × 2	**Step 1** Energy = power × time.
= 4.8 kWh	The dryer uses 4.8 kWh.
Price = 30 ÷ 100 = $0.30	**Step 2** Cents to dollars: **÷ 100**.
Cost = 4.8 × $0.30	**Step 3** Cost = energy × price.
= $1.44	Check: 5 kWh would cost $1.50.



multiply by the time, then by the price

WORKED EXAMPLE 11.4 — Working backwards

A 1200 W pool pump cost $3.00 to run at $0.25 per kWh. How many hours did it run?

WORKING	WHY
Energy = $3.00 ÷ $0.25	**Step 1** Energy = cost ÷ price.
= 12 kWh	The pump used 12 kWh.
Power = 1200 ÷ 1000 = 1.2 kW	**Step 2** W to kW: **÷ 1000**.
Time = 12 ÷ 1.2	**Step 3** Time = energy ÷ power.
= 10 h	Check: 1.2 × 10 × $0.25 = $3.00.



work backwards: divide instead of multiply

YOUR TURN 11

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 What is 2500 W in kilowatts (kW)? → Key idea

A. 0.25 kW **B.** 2.5 kW
C. 25 kW **D.** 2500 kW

2 How much energy does an 1800 W hair dryer use in 2 hours? → Example 11.1

A. 0.9 kWh **B.** 3.6 kWh
C. 900 kWh **D.** 3600 kWh

3 A 1.2 kW dishwasher runs for 5 h at 25 c per kWh. What is the cost?
Hint: 25 c = $0.25. → Example 11.3

A. $0.30 **B.** $1.25
C. $1.50 **D.** $6.00

Short answer show your working in the box

1 Find the energy each appliance uses, in kWh. *Hint: change W to kW first (÷ 1000).* → Example 11.1

(a) a 2.5 kW oven for 2 h (b) a 200 W TV for 5 h (c) a 50 W laptop for 12 h

(a)	(b)	(c)

2 Find the energy each appliance uses, in Wh. *Hint: W × h gives Wh, and half an hour = 0.5 h.* → Example 11.2

(a) a 15 W shaver for 2 h (b) a 900 W iron for half an hour (c) Write (b) in kWh.

(a)	(b)	(c)

3 At 20 c per kWh, find the cost of using each amount of energy. *Hint: 20 c = $0.20.* → Example 11.3

(a) 4 kWh (b) 25 kWh (c) 0.5 kWh

(a)	(b)	(c)

4 A 250 W slow cooker runs for 10 hours. Electricity costs 40 c per kWh. Find the cost. *Hint: kW first, then kWh.* → Examples 11.1 and 11.3

5 A 600 W vacuum cleaner cost $2.70 to run. Electricity costs $0.30 per kWh. For how many hours was it used? *Hint: energy = $2.70 ÷ $0.30.* → Example 11.4

Extended response answer each part in order

1 Jai's 400 W gaming PC runs for 5 hours a day. Electricity costs 30 c per kWh. → Examples 11.1 and 11.3

(a) Write the power of the PC in kW.

(b) How much energy does the PC use in one day, in kWh?

(c) How much does it cost to run the PC for one day? *Hint: 30 c = $0.30.*

(d) How much does it cost for one week (7 days)? *Hint: 7 × your answer to (c).*

(a)	(b)	(c)	(d)

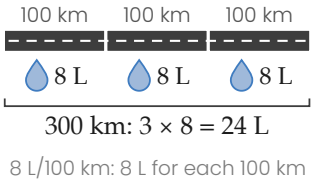
12 Fuel consumption

Key idea: Fuel consumption • Textbook 1J • MST-12-S2-05

Key idea Fuel consumption

Fuel consumption is a rate. It tells you how much fuel a car uses, usually in **litres per 100 kilometres** (L/100 km).

- **8 L/100 km** means the car uses 8 litres of fuel for **every** 100 km it travels.
- A **lower** number is better. It means the car uses **less** fuel, so it is more **fuel efficient** and cheaper to run.



$$\textbf{Fuel used (L)} = \frac{\text{distance (km)}}{100} \times \text{rate (L/100 km)} \qquad \textbf{Rate (L/100 km)} = \frac{\text{litres}}{\text{km}} \times 100$$

L/100 km or km/L?

Rate	Tells you	Better car
L/100 km	litres for 100 km	**lower** number
km/L	km on 1 litre	**higher** number

New cars in Australia show their fuel consumption in L/100 km on a windscreen label.

Cost of fuel

- **$1.80/L** means $1.80 for **each** litre.
- **Cost = litres × price per litre**
- The car in the picture uses 24 L. At $1.80/L, the fuel costs 24 × $1.80 = $43.20.

How far on this much fuel?

Use the unitary method. **Divide** to find the km for 1 litre, then **multiply**.

650 km on 50 L → 1 L: 650 ÷ 50 = 13 km → 20 L: 20 × 13 = **260 km**

CHECKLIST · SUCCESS CRITERIA Finished? Check the answers, then rate yourself: ○○○ not yet / getting there / got it

- ▸ *I can* calculate the fuel used on a trip, and the cost of the fuel. ○○○
- ▸ *I can* calculate the fuel consumption rate of a motor vehicle. ○○○

WORKED EXAMPLE 12.1 Fuel used for a trip

A car uses 9 L/100 km. How much fuel does it use on each trip? (a) 200 km (b) 350 km (c) 80 km

	WORKING	WHY
(a)	Fuel used = 200 ÷ 100 × 9	Fuel used = **distance ÷ 100** × rate.
	= 2 × 9	200 km is **2** lots of 100 km.
	= 18 L	The car uses 9 L for **each** 100 km.
(b)	Fuel used = 350 ÷ 100 × 9	Use the same rule with the new distance.
	= 3.5 × 9	350 km is **3.5** lots of 100 km.
	= 31.5 L	Check: between 3 × 9 = 27 and 4 × 9 = 36.
(c)	Fuel used = 80 ÷ 100 × 9	This trip is **shorter** than 100 km.
	= 0.8 × 9	80 km is **0.8** of 100 km.
	= 7.2 L	Less than 100 km, so less than 9 L.

WORKED EXAMPLE 12.2 — Finding the fuel consumption rate

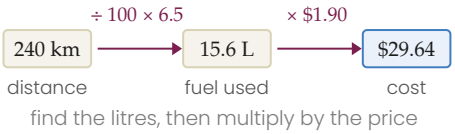
(a) A car uses 42 L of petrol to travel 600 km. Find its fuel consumption in L/100 km.
(b) A ute uses 66 L to travel 550 km. Find its fuel consumption. (c) Which is more fuel efficient?

WORKING	WHY
(a) Rate = 42 ÷ 600 × 100	Rate = litres ÷ km × 100. The **litres** go first.
= 0.07 × 100	Use your calculator: 42 ÷ 600 = 0.07
= 7 L/100 km	Check: 600 km is 6 lots of 100 km, and 6 × 7 = 42 L.
(b) Rate = 66 ÷ 550 × 100	× 100 gives the litres for **100** km, not 1 km.
= 0.12 × 100	Use your calculator: 66 ÷ 550 = 0.12
= 12 L/100 km	Check: 5.5 × 12 = 66 L.
(c) 7 is lower than 12	Compare the rates from (a) and (b).
The car	A **lower** rate means **less** fuel for each 100 km.

WORKED EXAMPLE 12.3 — The cost of fuel for a trip

Sam's car uses 6.5 L/100 km. Petrol costs $1.90 per litre. Find the cost of the petrol for a 240 km trip.

WORKING	WHY
Fuel used = 240 ÷ 100 × 6.5	**Step 1** Find the litres for the trip.
= 2.4 × 6.5	240 km is 2.4 lots of 100 km.
= 15.6 L	The trip uses 15.6 litres.
Cost = 15.6 × $1.90	**Step 2** Cost = litres × price.
= $29.64	Check: 16 L at $2 would be $32.

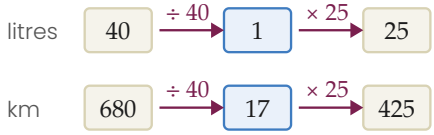


find the litres, then multiply by the price

WORKED EXAMPLE 12.4 — How far on this much fuel?

A small car travels 680 km on 40 L of petrol. How far can it travel on 25 L?

WORKING	WHY
40 L: 680 km	Write what you know, km on the **right**.
1 L: 680 ÷ 40 = 17 km	**Divide** by 40 to go down to **1** L (17 km/L).
25 L: 25 × 17	**Multiply** by 25 to go up to 25 litres.
= 425 km	Check: less fuel, so less than 680 km.



down to 1 litre, then up to 25 litres

YOUR TURN 12

Stuck? Each question shows you where to look.

Multiple choice circle the best answer

1 Which rate (in L/100 km) is the most fuel efficient?
→ Key idea

A. 9.2 **B.** 11.5 **C.** 7.4 **D.** 6.8

2 A car uses 6 L/100 km. How much fuel does it use for 450 km? → Example 12.1

A. 2.7 L **B.** 27 L
C. 75 L **D.** 2700 L

3 A van travels 360 km on 54 L of petrol. What is its rate in L/100 km?
Hint: litres ÷ km × 100.
→ Example 12.2

A. 0.15 **B.** 6.7 **C.** 15 **D.** 194.4

Short answer show your working in the box

1 A car uses 11 L/100 km. How much fuel does it use on each trip? *Hint: distance ÷ 100 × 11.* → Example 12.1

(a) 500 km (b) 250 km (c) 60 km

(a)	(b)	(c)

2 Find the fuel consumption of each car, in L/100 km. *Hint: litres ÷ km × 100.* → Example 12.2

(a) Kim's car: 20 L for 250 km (b) Leo's car: 39 L for 300 km (c) Whose car is more efficient?

(a)	(b)	(c)

3 Petrol costs $1.75 per litre. Find the cost of each amount of petrol. *Hint: cost = litres × $1.75.* → Example 12.3

(a) 20 L (b) 46 L (c) 32.4 L

(a)	(b)	(c)

4 Ben's car uses 5.5 L/100 km. He drives 800 km. Petrol costs $1.70 per litre.
Hint: litres first. → Example 12.3

(a) How many litres? (b) What is the cost?

(a)	(b)

5 A car travels 960 km on 80 L of petrol.
Hint: divide to go down to 1 L. → Example 12.4

(a) How far on 1 L? (b) How far on 50 L?

(a)	(b)

Extended response answer each part in order

1 Rata drives 150 km from Sydney to Newcastle. His car uses 14 L/100 km. Petrol costs $1.60 per litre. → Examples 12.1, 12.2 and 12.3

(a) How many litres of petrol does the trip use?

(b) How much does the petrol cost? *Hint: litres × $1.60.*

(c) His friend's hybrid car uses 6 L for the same trip. Find its fuel consumption in L/100 km.

(d) Which car is more fuel efficient? *Hint: the lower rate uses less fuel.*

(a)	(b)	(c)	(d)

R Chapter review

Ratios and rates: all key ideas • MST-12-S2-05

BEFORE YOU START

- Look back at your **checklist ratings** in Lessons 1 to 12. Any criterion you rated ○ **not yet** or ○ **getting there** is a good place to start.
- **Show all your working, include units** in your answers, and **check** that each answer is sensible.
- Stuck? Every question shows the lesson to look back at, like this: → Lesson 4 The key ideas on page 3 sum up every lesson, and the contents on page 2 shows where each lesson starts.

Multiple choice circle the best answer

1 At an animal shelter, the ratio of cats to dogs is 4 : 7. There are 56 dogs. How many cats are there? → Lesson 1

A. 8 **B.** 14 **C.** 32 **D.** 98

2 A 900 g bag of flour is divided into two containers in the ratio 5 : 4. What is the mass of the smaller amount? → Lesson 3

A. 100 g **B.** 400 g
C. 500 g **D.** 720 g

3 A scale drawing has a scale of 1 : 25. A line on the drawing is 40 mm long. What is the actual length? → Lesson 4

A. 1.6 mm **B.** 40 mm
C. 65 mm **D.** 1000 mm

4 The scale on a map is 1 mm = 40 m. Two places are 800 m apart. How far apart are they on the map? → Lesson 4

A. 20 mm **B.** 200 mm
C. 840 mm **D.** 32 000 mm

5 A dog groomer washes 1 dog every 45 minutes. How many hours does it take to wash 8 dogs? → Lesson 9

A. 3.6 **B.** 6 **C.** 36 **D.** 360

6 A tap fills a 12 L watering can in 30 seconds. What is the rate of flow in litres per minute? *Hint: 30 s is half a minute.* → Lesson 8

A. 0.4 L/min **B.** 6 L/min
C. 12 L/min **D.** 24 L/min

7 Which is the **slowest** speed? *Hint: change each to km/h.* → Lessons 8 and 10

A. 250 m/min **B.** 20 km/h
C. 10 m/s **D.** 30 000 m/h

8 How much energy does an 800 W heater use in 5 hours? → Lesson 11

A. 0.16 kWh **B.** 0.8 kWh
C. 4 kWh **D.** 4000 kWh

9 Ravi uses a 4 kW air conditioner for 2 hours. He pays 25 c per kWh. What is the cost? → Lesson 11

A. $0.50 **B.** $1.00
C. $2.00 **D.** $8.00

Multiple choice continued

10 A motorbike travels 300 km using 15 L of petrol. What is its fuel consumption, in L/100 km? → Lesson 12

A. 0.05 **B.** 5 **C.** 20 **D.** 45

FINISHED THE MULTIPLE CHOICE?

Check your answers at the back of the workbook before you start the short answer. Got one wrong? Go back to the lesson shown, then try it again.

My score: _____ out of 10

Short answer show your working in the box

1 Write each ratio in simplest form. → Lesson 1

(a) 45 : 20 (b) 16 : 24 : 40 (c) 3.5 : 1.5

(a) (b) (c)

2 A ranger catches 90 turtles in a lake, tags them and lets them go. A month later she catches 60 turtles, and 15 of them have tags. → Lesson 2

(a) What fraction of the 2nd sample has tags? Simplify it.

(b) Estimate the number of turtles in the lake.

(a) (b)

3 Priya and Sam share the profits from their market stall in the ratio 3 : 5. → Lesson 3

(a) One week the profit is $640. How much does Priya get?

(b) The next week the profit is $1200. How much does Sam get?

(a) (b)

4 Write each scale as a ratio in the form $1 : x$. → Lesson 4

(a) 1 cm to 5 m

(b) 1 mm to 8 cm

(c) 1 m to 2 km

(a)	(b)	(c)

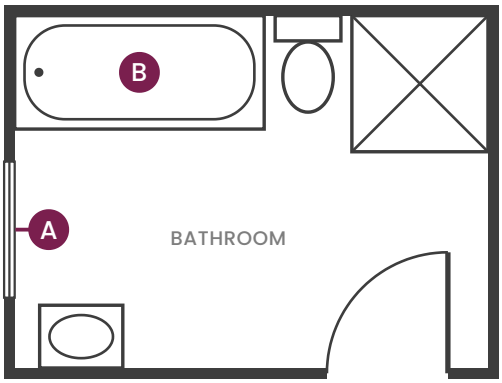
5 A map has the scale 1 cm = 4 km. Two towns are 65 mm apart on the map. How far apart are they in real life, in km? → Lesson 4

6 This floor plan of a bathroom is drawn at a scale of 1 : 50. → Lesson 5

(a) Name the symbol labelled A.

(b) Name the symbol labelled B.

(c) On the plan, the bathroom is 64 mm long. How long is the real bathroom, in metres?



floor plan, scale 1:50

(a)	(b)	(c)

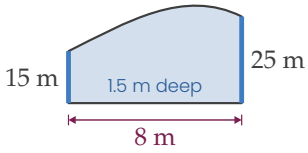
7 A lake in a park has one curved side. The water is 1.5 m deep everywhere. → Lessons 6 and 7

(a) Use the trapezoidal rule to estimate the area of the lake.

(b) Find the volume of water in the lake, in m^3.

(c) Change your answer to (b) into litres.

(d) 12 mm of rain falls on the lake. Find the volume of the rain, in m^3. *Hint: change 12 mm to metres first.*



plan of the lake (not to scale)

(a)	(b)	(c)	(d)

8 Convert each rate to the units shown. → Lesson 8

(a) 7200 m/h to m/min

(b) 6 cm/s to mm/s

(c) \$30/kg to \$/g

(a)	(b)	(c)

9 A 3 kg bag of oranges costs $5.40. → Lesson 9

(a) Find the cost of 1 kg.

(b) Find the cost of 8 kg.

(c) A 5 kg bag costs $8.50. Which bag is the better buy?

(a)	(b)	(c)

10 A delivery van travels at 70 km/h. → Lesson 10

(a) How far does it travel in 1.5 hours?

(b) How long does it take to travel 245 km? Give the time in hours and minutes.

(a)	(b)

11 An electric car is charged at home for 5 hours. The charger has a power of 7000 W. Electricity costs 20 c per kWh. → Lesson 11

(a) Write 7000 W in kW.

(b) How much energy is used, in kWh?

(c) How much does it cost to charge the car?

(a)	(b)	(c)

12 A car travels 560 km on 40 L of petrol. → Lesson 12

(a) How far does it travel on 1 L?

(b) How far can it travel on 30 L?

(a)	(b)

Extended response answer each part in order

1 There are 600 Year 7 students and 400 Year 8 students going to the swimming carnival. The school hires 20 buses. The buses are shared in the ratio Year 7 : Year 8. → Lesson 3

(a) Write the ratio Year 7 : Year 8 in simplest form. *Hint: divide both numbers by 200.*

(b) How many parts are there in total?

(c) How many buses is one part?

(d) How many buses are for Year 7? How many are for Year 8?

(a)	(b)	(c)	(d)

2 Two hoses are used to fill a 240 L paddling pool. Hose A on its own fills the pool in 12 minutes. Hose B on its own fills it in 24 minutes. → Lessons 8 and 9

(a) How many litres per minute does hose A pour in? *Hint: 240 L in 12 minutes.*

(b) How many litres per minute does hose B pour in?

(c) Both hoses are turned on together. How many litres per minute do they pour in altogether?

(d) How long do both hoses take to fill the pool? *Hint: 240 L ÷ your answer to (c).*

(a)	(b)	(c)	(d)

FINISHED THE REVIEW?

1. Check your answers at the back of the workbook. Fix any mistakes in a different colour.
2. For each question you got wrong, go back to the lesson shown and read the worked examples again.
3. Rate yourself again on the checklist in each lesson.

Answers

Lesson 1 Ratios

MULTIPLE CHOICE

1 A **2** B

3 D (1 part = 630 ÷ 9 = 70, adults = 5 × 70 = 350)

SHORT ANSWER

1 (a) 2 : 3 (b) 2 : 3 : 4 (c) 3 : 5

2 (a) 5 : 2 (b) 1 : 0.4 (c) 2.5 : 1

3 (a) $a = 10$ (b) $a = 32$ (c) $b = 20$

4 1 part = 4 ÷ 2 = 2, sand = 5 × 2 = 10 buckets

5 1 part = 176 ÷ 8 = 22 cm, Willow = 7 × 22 = 154 cm

EXTENDED RESPONSE

1 (a) 5 : 3
(b) 600 ÷ 3 = 200 mL
(c) 5 × 200 = 1000 mL
(d) 600 + 1000 = 1600 mL

Lesson 2 Capture–recapture

MULTIPLE CHOICE

1 C

2 B (10 out of 40 is 1 in 4, 25 × 4 = 100)

3 A (60 out of 600 is 1 in 10, 30 ÷ 10 = 3)

SHORT ANSWER

1 (a) 60 (b) 120 (c) 800

2 (a) $\frac{16}{40} = \frac{2}{5}$ (b) 50 × 40 ÷ 16 = 125 carp

3 200 × 50 ÷ 20 = 500 fish

4 $\frac{80}{400} = \frac{1}{5}$, 30 ÷ 5 = 6 possums

5 (a) fewer (b) too big

EXTENDED RESPONSE

1 (a) $\frac{6}{30} = \frac{1}{5}$
(b) 40 × 5 = 200 kangaroos
(c) So the tagged kangaroos can mix in with the others.
(d) Any one: no kangaroos are born, die, arrive or leave; the tags stay on; every kangaroo has the same chance of being caught.

Lesson 3 Dividing a quantity in a given ratio

MULTIPLE CHOICE

1 B **2** A (360 ÷ 12 = 30, 5 × 30 = 150 g)

3 D ($90 ÷ 9 = $10, 5 × $10 = $50)

SHORT ANSWER

1 (a) $240 and $160
(b) $320 and $80 (c) $120 and $280

2 (a) $700 ÷ 7 × 3 = $300
(b) $1400 ÷ 7 × 4 = $800

3 1 part = $100 ÷ 10 = $10, Jack $20, Kate $30, Leo $50

4 1 part = 35 ÷ 7 = 5 kg, compost = 3 × 5 = 15 kg

5 1 part = 90 ÷ 9 = 10 cm, longer = 7 × 10 = 70 cm

EXTENDED RESPONSE

1 (a) 160 : 60 = 8 : 3
(b) 8 + 3 = 11
(c) 22 ÷ 11 = 2 lifeboats
(d) passengers 8 × 2 = 16, crew 3 × 2 = 6 (16 + 6 = 22)

Lesson 4 Scale drawings

MULTIPLE CHOICE

1 C **2** D (20 × 20 = 400 mm)

3 A (600 ÷ 150 = 4 mm)

SHORT ANSWER

1 (a) 160 m (b) 300 m (c) 20 m

2 (a) 2000 cm = 20 m
(b) 8000 cm = 80 m (c) 600 cm = 6 m

3 (a) 1 : 300 (b) 1 : 60 (c) 1 : 2500

4 3 m = 300 cm, 300 ÷ 50 = 6 cm

5 46 mm = 4.6 cm, 4.6 × 5 = 23 km

EXTENDED RESPONSE

1 (a) 6 × 50 000 = 300 000 cm
(b) 300 000 ÷ 100 000 = 3 km
(c) 2.5 × 100 000 = 250 000 cm
(d) 250 000 ÷ 50 000 = 5 cm

Lesson 5 Plans and elevations

MULTIPLE CHOICE

1 A **2** B

3 B (1 × 100 = 100 mm = 10 cm)

SHORT ANSWER

1 (a) plan: 3 squares in a row 

(b) front: 3 squares with 1 on top of the middle 

(c) side: 2 squares, one on top of the other 

2 (a) 3000 mm = 3 m

(b) 4500 mm = 4.5 m

(c) 6200 mm = 6.2 m

3 9 × 100 = 900 mm,
900 ÷ 10 = 90 cm

4 40 mm : 2000 mm = 40 : 2000 = 1 : 50

EXTENDED RESPONSE

1 (a) water closet: a toilet

(b) a long rectangle with a rounded shape inside (and a dot for the plug)

(c) 30 : 3000 = 1 : 100

(d) 20 × 100 = 2000 mm = 2 m

Lesson 6 Perimeter and area of land

MULTIPLE CHOICE

1 B (6 × 20 = 120 m)

2 C (8 + 4 ÷ 2 = 10 squares,
10 × 100 = 1000 m^2)

3 D (5 × (18 + 22) = 5 × 40 = 200 m^2)

SHORT ANSWER

1 (a) 4 × 20 = 80 m, 3 × 20 = 60 m

(b) 80 + 60 + 80 + 60 = 280 m

(c) 80 × 60 = 4800 m^2

2 (a) 10 (b) 4

(c) 10 + 4 ÷ 2 = 12 squares,
12 × 100 = 1200 m^2

3 (a) π × 6 ÷ 2 = 9.4 m

(b) 9.4 + 6 = 15.4 m

(c) π × 3^2 ÷ 2 = 14.1 m^2

4 15 × (25 + 35) = 15 × 60 = 900 m^2

5 5 × (12 + 18) + 5 × (18 + 14)
= 150 + 160 = 310 m^2

EXTENDED RESPONSE

1 (a) π × 8 ÷ 2 = 12.6 m

(b) 8 + 4 + 4 + 12.6 = 28.6 m

(c) 2 × 12 + 2 × 12 = 24 + 24 = 48 m^2

Lesson 7 Volume of water

MULTIPLE CHOICE

1 C **2** B (25 ÷ 1000 = 0.025 m)

3 D (40 × 0.5 = 20 m^3)

SHORT ANSWER

1 (a) 5000 L (b) 2.5 m^3 (c) 0.04 m

2 (a) 8 × 5 = 40 m^2

(b) 40 × 2 = 80 m^3 (c) 80 000 L

3 (a) 3 × (20 + 40) = 180 m^2

(b) 180 × 2.5 = 450 m^3

4 (a) 9 ÷ 1000 = 0.009 m

(b) 180 × 0.009 = 1.62 m^3 (c) 1620 L

EXTENDED RESPONSE

1 (a) 48 × 1.5 = 72 m^3

(b) 72 × 1000 = 72 000 L

(c) underestimate: the dashed lines are inside the curved edge, so the rule leaves out some of the pond.

Lesson 8 Rates

MULTIPLE CHOICE

1 A ($96 ÷ 4 = $24/h)

2 C (30 × 60 = 1800 L/h)

3 B (6000 m/h = 6 km/h;
2000 m/min = 120 km/h)

SHORT ANSWER

1 (a) $6/kg (b) 7 L/min

(c) 60 km/h

2 (a) 5 × 60 = 300 bottles/min

(b) 14 400 ÷ 60 = 240 m/min

(c) 50 × 60 = 3000 mL/h

3 (a) 8 × 10 = 80 mm/s

(b) 7000 ÷ 1000 = 7 km/h

(c) $15 ÷ 1000 = $0.015/g

4 30 × 60 = 1800 mL/h,
1800 ÷ 1000 = 1.8 L/h

5 120 × 60 = 7200 cm/min,
7200 × 10 = 72 000 mm/min

EXTENDED RESPONSE

1 (a) 1200 ÷ 4 = 300 bottles/min

(b) 300 × 60 = 18 000 bottles/h

(c) 300 ÷ 60 = 5 bottles/s

(d) the new machine
(6 bottles/s is more than 5 bottles/s)

Lesson 9 The unitary method and best buys

MULTIPLE CHOICE

1 C (1 kg: \$10 ÷ 4 = \$2.50, 6 kg: 6 × \$2.50 = \$15)

2 D (per L: \$2.50, \$2.30, \$2.40, \$2.10)

3 A (24 × 25 = 600 min, 600 ÷ 60 = 10 h)

SHORT ANSWER

1 (a) \$9.20 ÷ 5 = \$1.84
(b) 3 × \$1.84 = \$5.52
(c) 10 × \$1.84 = \$18.40

2 (a) \$580 ÷ 20 = \$29
(b) 35 × \$29 = \$1015

3 1 person: 3 ÷ 6 = 0.5 egg, 14 people: 14 × 0.5 = 7 eggs

4 15 × 12 = 180 min, 180 ÷ 60 = 3 h

5 (a) \$3.25 ÷ 1.25 = \$2.60 per L
(b) \$4.20 ÷ 2 = \$2.10 per L
(c) the 2 L bottle

EXTENDED RESPONSE

1 (a) 200 g: 2 lots, 500 g: 5 lots
(b) \$9.00 ÷ 2 = \$4.50
(c) \$20.00 ÷ 5 = \$4.00
(d) the 500 g jar (\$4.00 per 100 g)

Lesson 10 Speed and distance–time graphs

MULTIPLE CHOICE

1 B (210 ÷ 3 = 70 km/h)

2 C (a flat line means stopped)

3 C (5 ÷ 0.5 = 10 km/h)

SHORT ANSWER

1 (a) 42 ÷ 3 = 14 km/h
(b) 650 × 2 = 1300 km
(c) 150 ÷ 30 = 5 h

2 (a) 4 ÷ 0.25 = 16 km/h
(b) 21 ÷ 0.75 = 28 km/h
(c) 135 ÷ 60 = 2.25 h = 2 h 15 min

3 100 × 3.5 = 350 km

4 (a) 90 ÷ 3.6 = 25 m/s
(b) 72 ÷ 3.6 = 20 m/s

5 (a) 36 ÷ 3.6 = 10 m/s
(b) 400 ÷ 10 = 40 s
(c) yes: 40 s is less than 48.63 s

EXTENDED RESPONSE

1 (a) 30 ÷ 2 = 15 km/h
(b) 30 ÷ 1 = 30 km/h
(c) 60 ÷ 5 = 12 km/h
(d) 5 − 2 = 3 h, 60 ÷ 3 = 20 km/h

Lesson 11 Energy

MULTIPLE CHOICE

1 B (2500 ÷ 1000 = 2.5 kW)

2 B (1800 W = 1.8 kW, 1.8 × 2 = 3.6 kWh)

3 C (1.2 × 5 = 6 kWh, 6 × \$0.25 = \$1.50)

SHORT ANSWER

1 (a) 2.5 × 2 = 5 kWh
(b) 0.2 × 5 = 1 kWh
(c) 0.05 × 12 = 0.6 kWh

2 (a) 15 × 2 = 30 Wh
(b) 900 × 0.5 = 450 Wh
(c) 450 ÷ 1000 = 0.45 kWh

3 (a) 4 × \$0.20 = \$0.80
(b) 25 × \$0.20 = \$5.00
(c) 0.5 × \$0.20 = \$0.10

4 250 W = 0.25 kW,
0.25 × 10 = 2.5 kWh,
2.5 × \$0.40 = \$1.00

5 \$2.70 ÷ \$0.30 = 9 kWh,
600 W = 0.6 kW, 9 ÷ 0.6 = 15 h

EXTENDED RESPONSE

1 (a) 400 ÷ 1000 = 0.4 kW
(b) 0.4 × 5 = 2 kWh
(c) 2 × \$0.30 = \$0.60
(d) 7 × \$0.60 = \$4.20

Lesson 12 Fuel consumption

MULTIPLE CHOICE

1 D (the lowest rate)

2 B (450 ÷ 100 × 6 = 27 L)

3 C (54 ÷ 360 × 100 = 15 L/100 km)

SHORT ANSWER

1 (a) 5 × 11 = 55 L
(b) 2.5 × 11 = 27.5 L
(c) 0.6 × 11 = 6.6 L

2 (a) 20 ÷ 250 × 100 = 8 L/100 km
(b) 39 ÷ 300 × 100 = 13 L/100 km
(c) Kim's car (8 is lower than 13)

3 (a) 20 × \$1.75 = \$35.00
(b) 46 × \$1.75 = \$80.50
(c) 32.4 × \$1.75 = \$56.70

4 (a) 800 ÷ 100 × 5.5 = 44 L
(b) 44 × \$1.70 = \$74.80

5 (a) 960 ÷ 80 = 12 km
(b) 50 × 12 = 600 km

EXTENDED RESPONSE

1 (a) 150 ÷ 100 × 14 = 21 L
(b) 21 × \$1.60 = \$33.60
(c) 6 ÷ 150 × 100 = 4 L/100 km
(d) the hybrid car (4 is lower than 14)

Chapter review

MULTIPLE CHOICE

1 C (56 ÷ 7 = 8, 4 × 8 = 32)

2 B (900 ÷ 9 = 100 g, 4 × 100 = 400 g)

3 D (40 × 25 = 1000 mm)

4 A (800 ÷ 40 = 20 mm)

5 B (8 × 45 = 360 min,
360 ÷ 60 = 6 h)

6 D (12 L in half a minute,
12 × 2 = 24 L/min)

7 A (250 m/min = 15 km/h,
10 m/s = 36 km/h,
30 000 m/h = 30 km/h)

8 C (800 W = 0.8 kW,
0.8 × 5 = 4 kWh)

9 C (4 × 2 = 8 kWh,
8 × $0.25 = $2.00)

10 B (15 ÷ 300 × 100 = 5 L/100 km)

SHORT ANSWER

1 (a) 9 : 4 (b) 2 : 3 : 5 (c) 35 : 15 = 7 : 3

2 (a) $\frac{15}{60} = \frac{1}{4}$
(b) 90 × 4 = 360 turtles

3 (a) $640 ÷ 8 × 3 = $240
(b) $1200 ÷ 8 × 5 = $750

4 (a) 1 : 500 (b) 1 : 80 (c) 1 : 2000

5 65 mm = 6.5 cm, 6.5 × 4 = 26 km

6 (a) window (b) bath
(c) 64 × 50 = 3200 mm = 3.2 m

7 (a) 4 × (15 + 25) = 160 m^2
(b) 160 × 1.5 = 240 m^3 (c) 240 000 L
(d) 12 mm = 0.012 m,
160 × 0.012 = 1.92 m^3

8 (a) 7200 ÷ 60 = 120 m/min
(b) 6 × 10 = 60 mm/s
(c) $30 ÷ 1000 = $0.03/g

9 (a) $5.40 ÷ 3 = $1.80
(b) 8 × $1.80 = $14.40 (c) the 5 kg bag
($8.50 ÷ 5 = $1.70 per kg)

10 (a) 70 × 1.5 = 105 km
(b) 245 ÷ 70 = 3.5 h = 3 h 30 min

11 (a) 7000 ÷ 1000 = 7 kW
(b) 7 × 5 = 35 kWh
(c) 35 × $0.20 = $7.00

12 (a) 560 ÷ 40 = 14 km
(b) 30 × 14 = 420 km

EXTENDED RESPONSE

1 (a) 600 : 400 = 3 : 2
(b) 3 + 2 = 5
(c) 20 ÷ 5 = 4 buses
(d) Year 7: 3 × 4 = 12 buses,
Year 8: 2 × 4 = 8 buses (12 + 8 = 20)

2 (a) 240 ÷ 12 = 20 L/min
(b) 240 ÷ 24 = 10 L/min
(c) 20 + 10 = 30 L/min
(d) 240 ÷ 30 = 8 min